Smart India Hackathon 2026

Software Edition — Problem Statements

Total Software Problem Statements: 172
Source: sih.gov.in/sih2026PS

Breakdown by Theme

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1. [SIH26001] AI-Based early warning and landslide Risk Monitoring System in NER

Organization: Ministry of Development of North Eastern Region (MDoNER) | **Theme:** Disaster Management | **Submission Deadline:** 20 September 2026

Background:
The North Eastern Region (NER) frequently faces landslides, flash floods, road blockages, and slope failures due to heavy rainfall, fragile terrain, and unplanned hill cutting. These incidents often disrupt connectivity, damage infrastructure, delay emergency response, and isolate remote villages for days. Currently, monitoring of vulnerable zones is mostly reactive and dependent on manual reporting. There is limited use of real-time predictive systems for identifying high-risk zones and issuing early warnings to authorities and local communities. With increasing climate vulnerability in the region, there is a need for an AI-enabled real-time monitoring and prediction system that can help authorities take preventive action before disasters occur.
Description:
This problem statement proposes the development of an AI-powered early warning and monitoring platform capable of …

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2. [SIH26002] AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER)

Organization: Ministry of Development of North Eastern Region (MDoNER) | **Theme:** Smart Automation | **Submission Deadline:** 20 September 2026

Background:
The North Eastern Region (NER) faces major logistics and accessibility challenges due to difficult terrain, extreme weather conditions, limited transport connectivity, and frequent road disruptions caused by landslides, floods, and infrastructure gaps. Transportation of essential goods such as medicines, food supplies, construction materials, and agricultural produce to remote districts often gets delayed, leading to supply shortages, increased costs, and disruption in public service delivery. Currently, there is no integrated intelligent platform that can provide real-time logistics visibility, route accessibility status, predictive disruption alerts, and optimized transportation planning for the region. To strengthen regional connectivity and support infrastructure-led development initiatives, there is a need for an AI-enabled logistics intelligence system tailored for the …

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3. [SIH26003] AI-Based Cognitive Gaming and Memory Assistance Platform for Elderly Dementia Patients in North Eastern Region (NER)

Organization: Ministry of Development of North Eastern Region (MDoNER) | **Theme:** Space Technology | **Submission Deadline:** 20 September 2026

Background:
The North Eastern Region (NER) is witnessing a gradual rise in age-related cognitive disorders such as dementia and memory loss among the elderly population. Many families in remote and rural areas face challenges in accessing specialized neurological care, cognitive therapy, and long-term elderly support services due to limited healthcare infrastructure and geographical barriers.
Elderly patients suffering from dementia often experience memory decline, confusion, anxiety, and social isolation, while caregivers face difficulties in continuous monitoring and engagement. There is limited availability of affordable and culturally inclusive digital therapeutic solutions tailored for elderly individuals in the North-Eastern Region.
To strengthen elderly healthcare and improve cognitive well-being, there is a need for an accessible, engaging, and AI-enabled cognitive gaming …

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4. [SIH26006] Development of an Intelligent Freight Forecasting Model for Optimized Vessel Chartering and Bulk Cargo Procurement from overseas to East Coast of India

Organization: Ministry of Steel | **Theme:** Transportation & Logistics | **Submission Deadline:** 20 September 2026

Background:
The current approach to vessel chartering for bulk cargo procurement to India's East Coast ports often involves daily market exploration, leading to reactive decision-making and likely missed opportunities for cost savings and efficiency. The highly volatile nature of global freight markets, coupled with varying supply and demand dynamics from key origins like Australia, the US, Mozambique, Russia and Indonesia, makes it challenging to identify optimal entry points for short-term or mid-term charter contracts. Furthermore, without a robust future forecasting mechanism, determining the most suitable vessel type (e.g., Handysize, Supramax, Panamax,Capesize) for specific cargo parcels and routes, while accounting for port infrastructure limitations at both origin and destination, results in suboptimal utilization and increased idle time. This manual, market-dependent approach …

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5. [SIH26009] Using AI/ML and Space Technology to Identify Manganese Reserves and Overcome Production Shortfalls.

Organization: Ministry of Steel | **Theme:** Smart Automation | **Submission Deadline:** 20 September 2026

Background: MOIL Limited is the largest producer of Manganese Ore in India. To meet future demand, it is important to accurately identify available reserves and avoid production shortfalls. At present, reserve estimation and production planning are mainly based on manual surveys, drilling results, and production records. These methods are time-consuming and sometimes lead to a mismatch between expected and actual ore production. Detailed Description: The challenge is to develop an AI/ML-based solution that uses geological data,historical production, equipment performance, and satellite/space technology inputs (such as rainfall, soil moisture, vegetation index, and land temperature) to:
• Identify and map manganese reserves more accurately using surface and sub-surface indicators.
• Predict shortfalls in production by analysing constraints like equipment downtime, weather conditions, or …

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6. [SIH26011] 3D ULPIN Generation and vertical Property Mapping SYstem

Organization: Ministry of Rural Development | **Theme:** Space Technology | **Submission Deadline:** 20 September 2026

Background:
With rapid urbanization and vertical growth of cities, conventional 2D land record Systems are becoming inadequate for managing modern urban properties.Existing land administration systems are primarily designed to identify surface-level land parcels and are unable to uniquely define ownership rights associated with multi- storey apartments, underground infrastructure, elevated transport corridors, parking spaces, air-rights, and subsurface utility networks.
Description:
The proposed solution Should develop an advanced 3D ULPIN(Unique Land Parcel ldentification Number) Generation and vertical Property Mapping system capable of creating unique spatial identities for:
. Surface land parcels . Multi-storey apartments . Underground infrastructure The system should integrate:
. Drone imagery . LiDAR/3D Point cloud data . GIS parcel layers . Building floor Plans . GNSS/CORS-based …

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7. [SIH26012] AI-Based Automated Urban Parcel Mapping and Cadastral Feature Extraction System using Drone Imagery

Organization: Ministry of Rural Development | **Theme:** Robotics and Drones | **Submission Deadline:** 20 September 2026

Background:
Accurate and up-to-date urban land records are essential for effective land governance, urban planning, taxation, infrastructure development, and delivery of citizen-centric services. At present, preparation of cadastral maps and delineation of urban parcel boundaries is largely dependent on manual interpretation of drone imagery and field-based Ground Truthing (GT) activities. The process is time- consuming, resource intensive, and requires extensive human intervention for extraction of parcel boundaries, building footprints, road networks, and other cadastral features.Further, dense urban settlements, irregular parcel geometries, encroachments,overlapping structures, narrow access roads, and mixed land-use patterns create significant challenges in preparation of accurate parcel maps. Manual digitization and validation of parcel boundaries often lead to delays in completion …

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8. [SIH26013] Automated Integration and Intelligent Harmonization of Multi-source Geospatial Data for urban Land Record Management.

Organization: Ministry of Rural Development | **Theme:** Disaster Management | **Submission Deadline:** 20 September 2026

Background: Urban land administration and cadastral management involve integration of multiple spatial and non-spatial datasets generated from various departments,agencies,and survey mechanisms. Under modern land governance programmes such as the NAKSHA Programme, large volumes of geospatial data are being generated through drone surveys, Orthorectified Imagery (ORI), DSM/DTM datasets, Ground Truthing (GT),GNSS surveys,municipal records,utility databases, and revenue land records.
At present,harmonization and integration of these datasets largely depend on manual GIS workflows,which are time-consuming and prone to errors.With increasing availability of AI, GeoAI, and automated spatial processing technologies, there is significant scope for development of an intelligent system capable of automatically integrating and synchronizing multi-source geospatial datasets with feature-extracted …

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9. [SIH26014] An Integrated GIS-based Digital Public Infrastructure for Land Governance

Organization: Ministry of Rural Development | **Theme:** Robotics and Drones | **Submission Deadline:** 20 September 2026

Background:
Land governance in India involves multiple institutions maintaining land-related information in fragmented and disconnected systems. Core datasets such as cadastral maps, Record of Rights (RoR), registration records, land use information, Master Plan, Building Permission, Restrictions, property taxation records, utility infrastructure, and other land-related databases are often managed Independently by different departments and agencies with limited interoperability. This results in duplication of effort, inconsistencies in records, delays in obtaining ownership information, lack of transparency in transactions, and inconvenience to citizens seeking land-related services.
The growing scale of urbanization, increasing land transactions, demand for efficient governance, and the need for transparent and citizen-centric public service delivery require a modern digital approach …

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10. [SIH26015] Application of Geospatial Techniques for visualization and analysis to interpret Geo-Coded Images to enhance watershed Development Outcomes.

Organization: Ministry of Rural Development | **Theme:** Disaster Management | **Submission Deadline:** 20 September 2026

Background:
Watershed development plays a vital role in sustainable management of land, water, and natural resources, particularly in rural and semi-arid regions of India. Effective watershed planning and monitoring require accurate spatial information on land use, drainage patterns, vegetation cover, soil moisture, water bodies, and changes occurring over time. Traditional monitoring approaches often rely on field surveys and manual reporting, which are time-consuming, resource-intensive, and limited in spatial coverage. In recent years, advancements in Geographic Information systems (GIS), Remote Sensing (RS), and geospatial technologies have created new opportunities for scientific watershed assessment and evidence-based decision- making.Geo-coded images, integrated with satellite-based spatial datasets, provide location- specific visual information that can significantly improve …

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11. [SIH26016] Real-Time National Land Acquisition & Management System for End-to-End Digital Monitoring and Decision Support

Organization: Ministry of Rural Development | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

Background:
Land acquisition is a critical component of infrastructure development and public welfare projects in India. It facilitates the implementation of highways, railways, industrial corridors, irrigation projects, urban development, renewable energy initiatives, and other strategic infrastructure. The process involves multiple stakeholders, including land requiring bodies, land acquiring authorities, district administrations, state governments, and central ministries.
Description of the Study:
Web-based National Land Acquisition & Management System that digitizes the complete land acquisition lifecycle-from project proposal submission to final possession of land. The proposed platform should provide standardized workflows for different stakeholders and enable seamless coordination among Central Ministries, State Governments, District Authorities, and Project Implementing …

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12. [SIH26017] Predictive Analytics System for Early Detection of Land Acquisition Delays

Organization: Ministry of Rural Development | **Theme:** Agriculture, FoodTech & Rural Development | **Submission Deadline:** 20 September 2026

Background:
Land acquisition is one of the most critical and time-sensitive phases of infrastructure development. Delays in acquiring land significantly impact the execution of national and state-level projects. The causes of land acquisition delays are multifaceted, including prolonged administrative approvals, legal disputes, delayed compensation disbursement, incomplete documentation, pending notifications, land ownership conflicts, rehabilitation and resettlement challenges, and inter-departmental coordination issues.
Description of the Study:
Develop an AI-powered Predictive Analytics System capable of identifying land acquisition projects that are at risk of delay by analyzing historical and real-time project data.
The proposed solution should utilize machine learning algorithms to study patterns from completed and ongoing land acquisition cases, considering parameters such as …

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13. [SIH26018] Intelligent Land Record Digitization and Validation System

Organization: Ministry of Rural Development | **Theme:** MedTech / BioTech / HealthTech | **Submission Deadline:** 20 September 2026

Background:
Land records form the backbone of land administration, property ownership, taxation, land acquisition, dispute resolution, and infrastructure planning. Across India, a significant portion of historical land records continues to exist in the form of handwritten registers, scanned documents, maps, cadastral records, and legacy PDF files maintained at various administrative levels.
An intelligent digitization system can significantly improve data quality while accelerating the modernization of India's land administration ecosystem.
Description of the Study:
Develop an AI-powered Intelligent Land Record Digitization and Validation System capable of automatically extracting structured information from scanned land records, handwritten documents, maps, and legacy PDF files.
The proposed solution should utilize advanced OCR, Computer Vision, and Natural Language Processing …

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14. [SIH26019] National Digital Platform for Research, Policy Innovation, and Evidence-Based Land Governance

Organization: Ministry of Rural Development | **Theme:** Blockchain & Cybersecurity | **Submission Deadline:** 20 September 2026

Background:
Land is a finite and strategic resource that underpins economic development, environmental sustainability, food security, urban expansion, and social equity. Effective land governance is therefore critical to achieving sustainable development goals and supporting India's rapidly evolving socio-economic landscape. However, the land administration ecosystem in India remains largely implementation-oriented, with limited institutional focus on applied research, policy experimentation, and evidence-based innovation.
Description:Develop a comprehensive National Digital Platform for Research and Policy Innovation that promotes applied research, policy experimentation, knowledge sharing, and evidence-based decision-making in land governance.
The platform should function as a centralized repository and collaborative ecosystem that integrates datasets, research publications, policy …

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15. [SIH26021] Honey Chain: A block chain-based system for honey traceability and smart beekeeping management.

Organization: Ministry of MSME | **Theme:** Smart Automation | **Submission Deadline:** 20 September 2026

Background:
KVIC’s Honey Mission supports rural beekeepers with bee boxes and extraction toolkits for livelihood promotion, but they still face challenges like counterfeit honey, low consumer trust, weak market linkages, and lack of traceability and advanced hive management support.
Hence, there is a need for an integrated block chain, AI, and IoT-based digital ecosystem to improve honey authenticity, traceability, productivity, and market credibility.
Description:
Develop 'Honey Chain,' a block chain-based honey traceability and smart beekeeping system with QR-code consumer verification, secure batch tracking, and AI-IoT features for disease detection, environmental monitoring, and productivity prediction to enhance authenticity, transparency, and market access for rural beekeepers.
Expected Solution:
• Develop a prototype block chain-based honey traceability and smart beekeeping …

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16. [SIH26023] AI-Powered Geological, Mining and other Reporting Solution for CMPDI/CIL subsidiaries

Organization: Ministry of Coal | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

Background:
CMPDI/CIL subsidiaries play a key role in providing geological and mining information to the Ministry of Coal and responding to parliamentary and high-priority administrative inquiries. These reports require compilation of data from scanned PDFs, digital documents, spreadsheets, images, and historical archives. The current workflow is largely manual, resulting in:
• High dependence on individual expertise
• Delay in generating reports and analytics
• Higher probability of manual errors
• Limited ability to quickly retrieve insights when required Objectives:
• Deploy an automated platform for AI-assisted geological, mining and any other production figures document processing and reporting.
• Enhance data validation, consistency, and traceability across historical and contemporary datasets.
• Build an efficient, scalable foundation for future digital transformation initiatives …

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17. [SIH26024] AI-Based Smart Governance and Compliance Monitoring System for Coal Mines

Organization: Ministry of Coal | **Theme:** Smart Automation | **Submission Deadline:** 20 September 2026

Background:
The Indian coal mining sector involves large-scale operations spread across multiple subsidiaries, mine sites, contractors, regulatory bodies, and field offices. Governance-related activities such as statutory compliance monitoring, inspection tracking, safety observations, production reporting, environmental monitoring, worker attendance, contract management, grievance handling, and regulatory reporting are often managed through fragmented systems, manual documentation, spreadsheets, and delayed reporting mechanisms.
This leads to challenges such as data inconsistency, delayed decision-making, limited transparency, compliance gaps, duplication of records, weak monitoring of field-level activities, and difficulty in obtaining real-time operational insights. With increasing focus on transparency, accountability, sustainability, and digital governance, there is a need for an …

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18. [SIH26027] AI-Powered Automatic Block Planning to Maximize Asset Availability for Train Operations on Indian Railways

Organization: Ministry of Railways | **Theme:** Transportation & Logistics | **Submission Deadline:** 20 September 2026

Background: Railway maintenance for fixed infrastructure of Engineering, Traction Distribution, and Signal & Telecommunication departments is currently planned independently. Each department requests maintenance blocks/disconnections via the BDMS system. This planning process is decentralized and manual. This often leads to inefficient block utilization, poor coordination, and suboptimal scheduling,which may reduce asset availability and impact train operations. Detailed Description: Maintenance data-such as defects and overdue tasks-is maintained separately in systems like Track Management System (TMS), Signalling Maintenance & Management System (SMMS), and Traction Distribution Management System (TDMS). Meanwhile, the Control Office Application (COA) manages block corridor availability. Without integration and coordinated scheduling, maintenance blocks/disconnections are not optimally …

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19. [SIH26028] Dynamic Forecast of Expected Time of Arrival (ETA) for Coaching Trains

Organization: Ministry of Railways | **Theme:** Disaster Management | **Submission Deadline:** 20 September 2026

Background: Accurate forecasting of the Expected Time of Arrival (ETA) for coaching trains is vital for improving passenger satisfaction and operational efficiency in Indian Railways. Currently, ETA is often estimated using static schedules, current delays and in-built recovery times, which may not reflect real-time ground realities such as speed restrictions, congestion, unscheduled stoppages or historical patterns. As a result, passengers, station staff, and downstream logistics services face uncertainty and planning difficulties. With the growing demand for real-time train information and forecast, there is a pressing need to shift towards a data-driven, dynamic ETA prediction system that continuously adapts to actual train running conditions. Detailed Description: Indian Railways operates a vast network of passenger trains across diverse geographies, weather conditions, and traffic …

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20. [SIH26031] Quality assessment and grading of onions are often subjective and vary across procurement centers, resulting in disputes and inconsistencies.

Organization: Ministry of Consumer Affairs, Food & Public Distribution | **Theme:** Fitness & Sports | **Submission Deadline:** 20 September 2026

Expected Solution: Develop an AI-based mobile application that:
• Uses image processing to assess onion quality.
• Identifies damaged, rotten, sprouted, or undersized onions.
• Estimates Grade A and URS percentages.
• Generates a digital quality report instantly.
• Reduces human bias and improves transparency.

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21. [SIH26032] Farmers often face long waiting times, lack of information regarding procurement schedules, and uncertainty about procurement status.

Organization: Ministry of Consumer Affairs, Food & Public Distribution | **Theme:** Heritage & Culture | **Submission Deadline:** 20 September 2026

Expected Solution: Develop a platform that:
• Enables farmer registration and slot booking.
• Provides real-time queue management.
• Sends SMS/app notifications.
• Tracks procurement and payment status.
• Reduces congestion and waiting time at procurement centres.

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22. [SIH26033] Multiple intermediaries reduce farmers earnings and increase consumer prices.

Organization: Ministry of Consumer Affairs, Food & Public Distribution | **Theme:** MedTech / BioTech / HealthTech | **Submission Deadline:** 20 September 2026

Expected Solution: Create a digital marketplace that:
• Connects farmers/FPOs directly with consumers and bulk buyers.
• Provides logistics support.
• Uses AI for demand forecasting and route optimization.
Benefits:
• Better prices for farmers.
• Lower prices for consumers.
• Reduced supply chain inefficiencies.

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23. [SIH26034] Software System to check compliance of Packaged Commodities under Legal Metrology(Packaged Commodities) Rules, 2011 by scanning products, images and labels.

Organization: Ministry of Consumer Affairs, Food & Public Distribution | **Theme:** Agriculture, FoodTech & Rural Development | **Submission Deadline:** 20 September 2026

Background:
Packaged commodities are widely sold through retail stores, supermarkets and e-commerce platforms across India. Under the Legal Metrology Act, 2009 and the Legal Metrology(Packaged Commodities) Rules, 2011, every packaged commodity is required to bear mandatory declarations such as name and address of manufacturer/packer/importer, net quantity, Maximum Retail Price (MRP), month and year of manufacture/packing/import,consumer care details and other prescribed declarations in a specified format and manner.These declarations are important for ensuring transparency, fair trade practices and consumer protection. However, due to the large volume and variety of packaged products available in the market, manual inspection and compliance checking by enforcement agencies becomes time-consuming and resource intensive. Non-compliance such as missing declarations, incorrect font sizes, …

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24. [SIH26035] Development of a Software Program/Application for Generation of Test Reports for Non-Automatic Weighing Instruments (NAWI) as per OIML Recommendation R- 76

Organization: Ministry of Consumer Affairs, Food & Public Distribution | **Theme:** Smart Vehicles | **Submission Deadline:** 20 September 2026

Background:
Non-Automatic Weighing Instruments (NAWIs), such as electronic weighing scales, platform scales and weighbridges, are widely used in trade, commerce, healthcare, agriculture and industry where accurate measurement is essential for fair transactions and consumer protection.Under the Legal Metrology Act, 2009 and the Legal Metrology (General) Rules, 2011, such instruments used for transaction and protection are required to conform to prescribed standards,obtain model approval, and undergo verification and stamping before being put into use.
For granting model approval, NAWIs are evaluated by designated laboratories in accordance with OIML Recommendation R 76 – Non-Automatic Weighing Instruments, which specifies internationally accepted technical, metrological and performance requirements. The evaluation involves various metrological and functional tests, the results of which …

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25. [SIH26036] Development of an Online Verification System for Weighing and Measuring Instruments

Organization: Ministry of Consumer Affairs, Food & Public Distribution | **Theme:** Transportation & Logistics | **Submission Deadline:** 20 September 2026

Background:
Under the Legal Metrology Act, 2009 and the Legal Metrology (General) Rules, 2011, every weighing and measuring instrument used in transaction or protection is required to be periodically verified and stamped before being put into use. Verification activities are carried out by Legal Metrology Officers (LMOs) of the State Legal Metrology Departments and Government Approved Test Centres (GATCs) notified by the Government.These verification activities presently involve substantial manual processes including submission of applications, scheduling of verification, recording of observations, issuance of verification certificates, maintenance of records and monitoring of validity periods. In many cases, records are maintained physically or through isolated local systems, resulting in delays,and difficulty in monitoring verification status across jurisdictions. There is also a need …

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26. [SIH26037] Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads

Organization: MathWorks | **Theme:** Robotics and Drones | **Submission Deadline:** 20 September 2026

Background:
Most autonomous driving systems are developed for roads with clear lane markings, standard signage, predictable traffic flow, and controlled intersections. Indian roads are often very different. Vehicles of many types share the same space, including cars, buses, trucks, auto-rickshaws, twowheelers, bicycles, pedestrians, pushcarts, and animals. Drivers and pedestrians may change direction suddenly, merge without signalling, drive against traffic, or cross at unmarked locations. In many areas, road edges are unclear, potholes are common, and formal lane discipline is limited. These conditions make it difficult for traditional path planning methods that depend on structured road geometry and predictable motion. India has a large and diverse road network that includes village roads, crowded market areas, urban intersections, and highways. To support the safe deployment of …

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27. [SIH26038] Explainable AI for Diabetic Retinopathy Screening in Rural India

Organization: MathWorks | **Theme:** Clean & Green Technology | **Submission Deadline:** 20 September 2026

Background:
India has over 77 million diabetic adults - the second highest globally. Diabetic Retinopathy (DR) affects ~18% of this population and is a leading cause of preventable blindness. Early screening can prevent90% of vision loss, but India has only ~1 ophthalmologist per 100,000 rural population, making mass manual screening infeasible. Existing AI solutions function as black boxes, lack clinical validation rigor, and fail with variable image quality from portable fundus cameras in field conditions. A robust, explainable, and validated screening system is essential for deployment in primary healthcare centres across rural India.
Description:
Design a MATLAB-based retinal image analysis pipeline for automated DR screening addressing real-world deployment challenges:
1. Image Quality Assessment and Enhancement: Automatically evaluate fundus images for adequacy (focus, …

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28. [SIH26041] AR-Based Vocational Training Simulator for Industrial Safety in Jharkhand's Mining & Manufacturing Sector

Organization: Governmcnt of Jharkhand | **Theme:** Blockchain & Cybersecurity | **Submission Deadline:** 20 September 2026

Background:
Jharkhand is India's leading mineral-producing state, with coal mines, steel plants, and mica processing units employing hundreds of thousands of workers many of them young tribal recruits with no prior industrial exposure. Classroom-based safety training using static manuals has documented retention rates below 20% after one week. Live drills are operationally disruptive, and VR headset simulators are inaccessible to small-scale mines and contract workers.. The DGMS, Dhanbad, recorded 48 fatal mine accidents in Jharkhand in 2022-23, a large share involving workers with under 30 days of orientation. The Factories Act, 1948 and Mines Act, 1952 mandate periodic safety certification, yet no standardised digital training platform exists in regional languages, and physical certificates have no mechanism to verify comprehension.
Description:
Design and develop a mobile AR-based …

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29. [SIH26042] AI-Powered Vernacular Pedagogy and Real-Time Translation Tool for Mother Tongue-Based Primary Education

Organization: Governmcnt of Jharkhand | **Theme:** Smart Education | **Submission Deadline:** 20 September 2026

Background:
Jharkhand's PALASH Mother Tongue-Based Multilingual Education (MTB-MLE) programme has demonstrated measurable improvements in foundational literacy among tribal children. However,scaling the programme is severely bottlenecked by a shortage of teachers proficient in tribal languages including Ho, Mundari, and Santhali -languages with limited digital NLP resources. The vast majority of teachers assigned to tribal-area primary schools are Hindi-medium trained and lack the linguistic tools to deliver mother-tongue-based instruction. Without a technology bridge, the pedagogical intent of MTB-MLE cannot be realised at scale, and children in over 5,000 tribal-area primary schools continue to receive instruction in a language they do not comprehend at home.
Description:
Develop an AI-assisted translation and curriculum-generation software suite that enables non-nativespeaking …

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30. [SIH26043] A digital platform to crowdsource societal challenges and facilitate collaborative problem solving through universities and industry partnerships

Organization: Governmcnt of Jharkhand | **Theme:** Disaster Management | **Submission Deadline:** 20 September 2026

Background:
Communities across Jharkhand encounter numerous local challenges related to education,healthcare, agriculture, water management, sanitation, environment, rural livelihoods,accessibility, urban infrastructure, and public service delivery. While citizens are often the first to identify these issues, there is currently no structured mechanism through which they can submit such problems for systematic evaluation and innovation-driven resolution.At the same time, Higher Education lnstitutions (HEIs) possess significant academic expertise,research capabilities, and a large pool of students capable of developing practical solutions.Industries and start-ups also have technical expertise, financial resources, and implementation capabilities that can complement academic research. However, collaboration among citizens,universities, and industry remains largely fragmented and …

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31. [SIH26044] Portal for Academia - Industry collaboration for Skill Mapping, Internships and Placement

Organization: Ministry of Ayush | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

Background:
A significant gap exists between the skills acquired in academic institutions and the competencies expected by industries. Students often struggle to identify the skills required for their desired career paths, while industries face challenges in finding candidates with the right skill sets. Similarly, academicians have limited visibility into industry internship opportunities that could help them gain practical exposure and align teaching with current industry practices. There is a need for a unified platform that connects students, industries, and academicians, enabling seamless collaboration and skill development.
Description:
The proposed solution is a centralized Academia–Industry Collaboration Portal that serves as a one-stop platform for students, industries, and academicians.
Key features include:
• Skill Assessment: Students complete a questionnaire to evaluate …

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32. [SIH26045] IP-SAKTI Sahayak a multilingual, RAG-based (source-cited) AI assistant for Intellectual Property and regulatory guidance in Ayurveda, across national and international regimes.

Organization: Ministry of Ayush | **Theme:** Toys & Games | **Submission Deadline:** 20 September 2026

Background:
Ayurveda rests on a vast corpus of codified and community-held traditional knowledge (TK) and on therapeutics derived from plant, microbial and animal sources. Protecting and commercialising an Ayurvedic product means navigating several overlapping regimes at once: patents, geographical indications (GI), trademarks, copyright, designs, trade secrets and plant-variety rights; the Access-and-Benefit-Sharing duties that flow from India's sovereignty over its biological resources; and the drug-regulatory framework that decides whether a formulation is a classical medicine, a proprietary medicine, a new drug, a phytopharmaceutical, a food or a cosmetic. Practitioners, researchers, AYUSH startups and MSMEs and cultivators routinely struggle with this. The result is twofold: legitimate Ayurvedic innovation is under-protected and under-commercialised, while India's traditional …

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33. [SIH26046] AIIA Clinical Trials Dashboard - a real-time, cloud-based, GCP-compliant Clinical Trial Management System (CTMS) for Ayurveda research, with CDISC/FHIR-interoperable data, role-based KPIs, and integrated ethics, regulatory (CTRI / NDCT Rules 2019) and pharmacovigilance tracking.

Organization: Ministry of Ayush | **Theme:** Space Technology | **Submission Deadline:** 20 September 2026

Background:
The All India Institute of Ayurveda (AIIA) conducts and coordinates a growing portfolio of clinical research in Ayurveda - interventional and observational studies, multi-centre trials - and, as the host of the National Pharmacovigilance Coordination Centre (NPvCC) for ASU&H drugs, it also anchors nationwide safety surveillance. This activity is governed by a demanding compliance framework: mandatory prospective registration in the Clinical Trials Registry – India (CTRI); the Good Clinical Practice guidelines for ASU medicine (GCP-ASU) and the ICMR National Ethical Guidelines; the New Drugs and Clinical Trials Rules, 2019 where applicable; Institutional Ethics Committee oversight; and timely Adverse-Event / Serious-Adverse-Event reporting. Yet study status, recruitment, milestones, data quality and safety signals are typically tracked across spreadsheets and disconnected …

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34. [SIH26047] Patient Case-Taking Software

Organization: Ministry of Ayush | **Theme:** Smart Automation | **Submission Deadline:** 20 September 2026

Background:
1.1 The Clinical History-Taking Bottleneck in Indian Hospitals History taking - the structured elicitation of a patient's presenting complaints, history of present illness, past medical and surgical history, drug and allergy history, family and personal history, and a review of systems - is the single most important diagnostic activity in clinical medicine. Classical teaching holds that a well-conducted history yields the correct diagnosis in 70–80% of cases, even before examination or investigation. Yet in India's overburdened public hospital outpatient departments (OPDs), the time available for this critical interaction has collapsed to unsustainable levels.
India operates one of the most patient-dense healthcare systems in the world. Tertiary government hospitals and apex institutions routinely register 4,000–10,000 OPD patients per day, with a doctor-to-patient …

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35. [SIH26051] Software Based Model Development for Design of Area Specific Shelter for Thermal Comfort Maintenance.

Organization: DRDO | **Theme:** Agriculture, FoodTech & Rural Development | **Submission Deadline:** 20 September 2026

• Background:
The ambient atmospheric condition affects the temperature inside the shelter and makes thermal management necessary for maintenance of temperature in the comfortable range. The existing shelters for any region are generally not designed as per the requirements of a particular region and hence not energy efficient thus demands external thermal comfort maintenance system. Area specific designed shelters looks smart and one time solution for thermal management as per the atmospheric condition of the region.
This model development project work is specifically conceptualised keeping in mind the tough climatic condition of High Altitude cold Region like Ladakh and can be used for studying the design requirements of other climatic region shelters as well.
• Description:
The Ladakh region is blessed with high solar energy irradiance (1900-2100 kwh/m2/year) along with long average …

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36. [SIH26053] Adaptive Variable Resolution 2.5D Lidar Mapping for Dynamic Environment Perception

Organization: DRDO | **Theme:** Transportation & Logistics | **Submission Deadline:** 20 September 2026

• Background:
Autonomous navigation depends on the ability of a vehicle to perceive its surroundings with high precision. While 3D Lidar point clouds provide rich spatial data, processing millions of points in real-time creates immense computational bottlenecks and memory latency. Conversely, standard 2D occupancy grids lose critical height information necessary for detecting curbs, potholes, or overhanging obstacles. To balance precision and performance, there is a need for a 'foveated' mapping approach-similar to human vision- where the immediate vicinity is rendered in high detail for safety, and distant areas are simplified to reduce the processing load.
• Description:
The goal is to build a deep learning pipeline that transforms raw Lidar point clouds into a variable resolution 2.5D grid (an elevation map with semantic layers). The system must perform three primary tasks:
1. …

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37. [SIH26054] AI-Enabled Real-Time Digital Twin System for Health Monitoring, Fault Prediction and Mission Reliability Enhancement of Aero Piston Engines used in MALE UAVs.

Organization: DRDO | **Theme:** Robotics and Drones | **Submission Deadline:** 20 September 2026

• Background:
Medium Altitude Long Endurance (MALE) UAV are increasingly being deployed for Long-duration intelligence, surveillance, reconnaissance (ISR).
Communication relay maritime surveillance and strategic defence missions Reliability and availability of propulsion systems are critical for mission success because piston-engine failures during flight may lead to mission abort, asset loss, or unsafe recovery conditions.
Conventional engine monitoring systems used in UAVs are primarily thresholdbased and reactive in nature. These systems generally indicate failures only after abnormality has already occurred. Present approaches also have limited capability to estimate remaining useful life (RUL) predict degradation trends, or simulate mission-wise engine behavior under varying environmental and operating conditions.
A Digital Twin (DT) framework for aero piston engines can …

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38. [SIH26055] Smart Scan strategy for Electronic Warfare

Organization: DRDO | **Theme:** Clean & Green Technology | **Submission Deadline:** 20 September 2026

Development of Smart Scan Strategy for Electronic Warfare in the absence of prior reliable intelligence of emitters and their operating characteristics.
• Background Detection of hostile communication or radar signals starts with search / scan of a wide frequency spectrum which covers relevant emitters. Sensors with typically high sensitivity but with at least an order lower instantaneous bandwidth compared to overall bandwidth of the system are used to maintain surveillance over the entire spectrum. This requires a receiver / receivers to sweep over frequency bands. Hitherto strategies based on pre mission data / prior data (Open loop) are used. Usually the first priority is to rapidly sweep the entire band with the best speed possible. Open loop strategies focus only on this requirement and may lose time to nonthreatening emitters by not giving time to new or threatening ones.
• …

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39. [SIH26056] Development of a Real-time Airfare Price Index for India through Automated Web Scraping of Airline and Online Travel Aggregator Portals for Augmentation of the Consumer Price Index (CPI).

Organization: MoSPI | **Theme:** Travel & Tourism | **Submission Deadline:** 20 September 2026

• Background The Consumer Price Index (CPI) released by the National Statistical Office (NSO), Ministry of Statistics and Programme Implementation (MoSPI), is the primary measure of retail inflation in India and is used by the Reserve Bank of India (RBI) for setting monetary policy under the flexible inflation-targeting framework. The current CPI framework, however, collects 'Transport and Communication' sub-group prices, including air travel fares, primarily through manual price-collection from a limited set of outlets and ticketing offices. With over 90% of domestic air tickets in India now sold online through airline websites and Online Travel Aggregators (OTAs) such as MakeMyTrip, Yatra, EaseMyTrip, Cleartrip, Ixigo and Goibibo, manual collection no longer captures the highly dynamic, route-specific, and time-sensitive pricing that Indian consumers actually face. Airfares in India …

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40. [SIH26057] AI-Powered Automated Underwater Marine Debris and Anomaly Detection System using Side-Scan Sonar Imagery

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Renewable / Sustainable Energy | **Submission Deadline:** 20 September 2026

• Background The accumulation of anthropogenic (man-made) debris in marine ecosystems poses a critical threat to global biodiversity. Among the most destructive types of pollution are ‘ghost nets’-abandoned, lost, or discarded fishing gear. These nets continuously trap and kill marine life,destroy coral reefs, and damage commercial vessel propellers. Because the ocean is vast and dark, marine conservationists and underwater technologists rely on Side Scan Sonar (SSS) instruments. These sensors are towed behind ships or mounted on Autonomous Underwater Vehicles (AUVs) to create detailed acoustic maps of the seafloor.However, manual inspection of thousands of kilometers of sonar logs is incredibly slow, tedious, and prone to human error. Debris can easily blend into natural geological features like rock formations, sand ripples, and marine ridges. Automating this process via computer …

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41. [SIH26059] AI-Enabled Antarctic Sea-Ice, Iceberg Trajectory, and Navigation Decision Support System

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Smart Education | **Submission Deadline:** 20 September 2026

Develop an AI/ML-enabled decision support platform capable of forecasting Antarctic sea-ice concentration, predicting iceberg trajectories, and identifying safe and fuel-efficient navigation routes for research vessels using satellite,oceanographic and meteorological datasets.

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42. [SIH26060] Digital Platform for efficient remote management of Indian Antarctic Research Stations

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Disaster Management | **Submission Deadline:** 20 September 2026

Develop a Digital Twin framework for Maitri and Bharati stations integrating infrastructure, energy, logistics and environmental monitoring for efficient remote management.

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43. [SIH26061] AI-Driven Smart Energy Management System for Polar Research Stations

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

Develop an intelligent energy-management system using AI for load forecasting, renewable energy integration and fuel optimization under extreme polar conditions.

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44. [SIH26062] Integrated Polar Expedition Logistics and Asset Management System

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Toys & Games | **Submission Deadline:** 20 September 2026

Develop a centralized digital platform for expedition planning, cargo tracking, inventory management, personnel movement and emergency response.

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45. [SIH26063] Integrated Polar Science Outreach, Knowledge Repository and Media Dissemination Portal

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Space Technology | **Submission Deadline:** 20 September 2026

Develop a comprehensive outreach portal that archives expedition reports, scientific datasets, publications, photographs, videos and institutional activities while generating content for websites and social media.

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46. [SIH26066] OceanEmbed - Satellite Embedding-Based Deep Learning Framework for Reconstruction of Subsurface Ocean Temperature from Surface Satellite Observations.

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Space Technology | **Submission Deadline:** 20 September 2026

• Background Subsurface ocean temperature is a fundamental variable for understanding ocean circulation,upper-ocean heat content, stratification, climate variability, air-sea interaction and marine ecosystems. Accurate representation of the vertical ocean temperature is essential for applications such as marine heatwave monitoring, fisheries, and data assimilation, etc. However, direct measurements of subsurface temperature remain sparse because they rely primarily on in-situ observing systems such as ARGO profiling floats, moored buoys, gliders, and ship observations. While these observations provide valuable vertical information, their spatial and temporal coverage is insufficient for generating continuous, basin-scale subsurface fields.In contrast, satellite observations provide continuous, large-scale monitoring of surface ocean conditions at relatively high spatial and temporal …

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47. [SIH26067] Develop a web-based interactive 3D visualization platform that integrates numerical ocean model outputs and in-situ observations.

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Smart Automation | **Submission Deadline:** 20 September 2026

• Background India's vast Exclusive Economic Zone (EEZ) and coastline demand continuous, high-resolution monitoring of ocean state variables. INCOIS routinely generates and archives large volumes of ocean model outputs - including three-dimensional fields of temperature, salinity, current vectors,chlorophyll, etc. - as well as real-time and delayed-mode observations from autonomous instruments such as Argo profiling floats and underwater Gliders. These datasets are stored in NetCDF and ASCII/text formats and span multiple depth levels, spatial grids, and time steps.Despite the richness of this data, no integrated, web-based 3D visualization platform currently exists that can simultaneously render model fields and in-situ instrument observations in a single interactive environment. Existing tools are either desktop-bound, support only 2D plan views, or lack the ability to co-visualize …

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48. [SIH26068] WeatherGPT: Conversational AI for Weather Forecasting, Alerts, and Climate Information

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Disaster Management | **Submission Deadline:** 20 September 2026

• Background Weather information is often distributed through multiple portals, bulletins, satellite products, and forecast systems, making it difficult for common users, researchers, disaster managers, and government agencies to quickly obtain actionable insights.
There is a need for an intelligent conversational platform that can provide real-time weather information, forecasts, warnings, climate analysis, and decision support in natural language.
• Objective Develop an AI-powered chatbot platform named WeatherGPT that integrates meteorological datasets, forecasting models, and disaster warning systems to provide accurate, contextual, and multilingual weather intelligence through conversational interfaces.
• Key Features 1. Real-time weather information retrieval.
2. Natural language querying for weather forecasts.
3. Integration with numerical weather prediction (NWP) models such as …

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49. [SIH26069] National Weather Big Data Analytics Platform

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Disaster Management | **Submission Deadline:** 20 September 2026

Design and develop a scalable National Weather Big Data Analytics Platform capable of collecting and processing real-time weather-related information for India from multiple internet-based sources including social media platforms, public datasets, websites, APIs, and citizen reports. The platform should automatically collect weather related posts and information tagged with #IMD and other relevant weather hashtags, along with metadata such as date & time, city, state, GPS location, photos, videos, and event category, and store the information in a centralized database.
The system should leverage big data technologies and open-source tools to support large-scale real-time data ingestion, processing, storage, and visualization.
Participants are encouraged to use machine learning and AI-based techniques to identify fake or misleading reports, verify untrusted sources, remove duplicate …

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50. [SIH26070] To develop an Artificial Intelligence (AI) / Machine Learning (ML) based system for identification, classification, and prediction of different tropical cyclone patterns using multi-source satellite data.

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Smart Education | **Submission Deadline:** 20 September 2026

To develop an Artificial Intelligence (AI) / Machine Learning (ML) based system for identification, classification, and prediction of different tropical cyclone patterns using multi-source satellite data.

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51. [SIH26071] AI/ML-Based Integrated heavy rainfall Early Warning and Inundation Prediction System using Satellite, Radar, observational Weather and numerical weather prediction model data.

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Disaster Management | **Submission Deadline:** 20 September 2026

AI/ML-Based Integrated heavy rainfall Early Warning and Inundation Prediction System using Satellite, Radar, observational Weather and numerical weather prediction model data.

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52. [SIH26072] AIML based Nowcasting of thunderstorm and lightning using atmospheric observation including multiple radars, satellite, lightning and model data.

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Disaster Management | **Submission Deadline:** 20 September 2026

AIML based Nowcasting of thunderstorm and lightning using atmospheric observation including multiple radars, satellite, lightning and model data.

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53. [SIH26073] AI/ML-Based Intelligent Anomaly Detection for Automatic Weather Stations (AWS)

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Disaster Management | **Submission Deadline:** 20 September 2026

• Title SkyGuard AI: Intelligent Real-Time Anomaly Detection System for Temperature, Pressure, and Humidity Sensors in Automatic Weather Stations
• Background Automatic Weather Stations (AWS) are critical components of modern meteorological observation networks. These stations continuously monitor atmospheric parameters and provide real-time data for weather forecasting, climate monitoring, disaster management, aviation, agriculture, and scientific research.However, AWS observations often contain anomalies caused by sensor malfunction,communication failures, calibration drift, power fluctuations, harsh environmental conditions, and data corruption.Erroneous observations can significantly impact weather forecasting accuracy and decision-making systems. Traditional threshold-based quality control methods are often insufficient for identifying complex or hidden anomalies in meteorological …

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54. [SIH26074] Downscaling of weather forecast from Block level to Panchayat level: Inferring high-resolution plots/ data/ information from low-resolution plot /data /information /variables for agro-meteorological advisory services.

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Disaster Management | **Submission Deadline:** 20 September 2026

Downscaling of weather forecast from Block level to Panchayat level: Inferring high-resolution plots/ data/ information from low-resolution plot /data /information /variables for agro-meteorological advisory services.

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*55. [SIH26075] Participants are invited to design and develop **CAPACITY CONNECT A Digital Capacity Building and Learning Management Portal** to support organizational training, competency development, and knowledge sharing through a centralized web-based platform.*

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Smart Education | **Submission Deadline:** 20 September 2026

The solution should include secure signup and login functionality with three user roles: Trainee, Trainer, and Admin. Trainees should be able to create professional profiles with qualifications, work experience, interests, skills, and certificates, enroll in courses, access learning resources, attempt subject-wise MCQ assessments, and provide feedback on courses and training content.Trainers should be able to manage their profiles, create questionnaires with deadlines, monitor trainee participation and performance, and upload recorded lectures, presentations, and study materials in a trainer library accessible to trainees.The Admin module should provide user approval and role management features along with dashboards for monitoring courses, enrollments, certifications,assessments, and participation statistics. Admins should also be able to publish notifications, announcements, …

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56. [SIH26076] Development of personalized homepage for 'Mausam' mobile application:

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

• Health-conscious users Highlight Air Quality Index (AQI), pollen count, UV index, and humidity levels to help users manage allergies, asthma, or skin sensitivity.
• Outdoor fitness enthusiasts Show sunrise/sunset times, 'best running hours,' wind speed, and heat alerts to optimize workout planning.
• Beachgoers & surfers Display sea conditions, tide timings, wave height, and water temperature for safe and enjoyable beach activities.
• Travelers Provide quick access to saved destinations, severe weather alerts for flights, and packing suggestions (e.g., 'Carry a raincoat in London').
• Parents & families Emphasize school commute conditions, rain alerts, and severe weather warnings to plan daily routines.
• Agriculture & gardeners Show soil moisture, rainfall predictions, frost alerts, and seasonal planting guidance.
• Commuters Integrate weather with traffic updates, visibility …

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57. [SIH26077] AI-Driven Hyper-Local Early Warning System for Severe Weather Nowcasting

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Disaster Management | **Submission Deadline:** 20 September 2026

• Problem Statement India is highly vulnerable to rapidly intensifying, localized extreme weather events such as cloudbursts, severe thunderstorms, and flash floods. Traditional physics-based Numerical Weather Prediction (NWP) models often suffer from computational latency and struggle to capture the rapid, small-scale atmospheric changes that preceded these events. There is a critical need for a real-time, hyper-local early warning system capable of 'nowcasting' severe weather 2 to 6 hours before impact, providing actionable lead time for disaster management.
• Proposed Solution We propose an advanced AI predictive engine designed for high-precision severe-weather nowcasting. Specifically, the system simultaneously predicts the onset of highly localized, rapidly intensifying events, namely severe thunderstorms, cloudbursts, and the subsequent flash floods,with an actionable lead time …

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58. [SIH26078] AI-Driven Spatio-Temporal Tracking of Extreme Weather Anomalies in Medium-Range Forecasts

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Disaster Management | **Submission Deadline:** 20 September 2026

• Problem Statement Identifying and tracking the exact geographic footprints of extreme weather anomalies (such as severe cyclones, heat domes, or cold waves) within massive global Numerical Weather Prediction (NWP) outputs is computationally intensive and heavily reliant on manual interpretation. In medium-range forecasting (3 to 10 days), atmospheric chaos renders traditional deterministic models highly uncertain.
Furthermore, standard deep learning models (like standard CNNs or U-Nets) suffer from spectral smoothing-they tend to 'average out' spatial data, which destroys the extreme amplitudes (the high-intensity peaks of rainfall or wind speed) that forecasters actually need to track. There is a critical gap between broad, coarse 12 km global ensemble datasets and localized, high-fidelity threat tracking.
• Proposed Solution We propose an automated, state-of-the-art AI tracking and …

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59. [SIH26079] AI-Based Forecast Bust Detection for Medium-Range Weather Forecasts

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Disaster Management | **Submission Deadline:** 20 September 2026

• Problem Statement Medium-range weather forecasts sometimes show large errors during rapidly evolving systems such as monsoon depressions, heavy rainfall events, western disturbances, cyclones, heat waves and break/active monsoon phases. Such forecast failures, or 'forecast busts', can affect operational decision-making.
• Challenge The challenge is to develop an AI/ML-based system that can identify regions and lead times where the forecast is likely to have high uncertainty or large error. The system should compare current NWP forecast patterns with historical forecast error behaviour and provide a forecast confidence indicator.
Expected Outcome - Description Forecast confidence map - Region-wise confidence for Day 1 to Day 10 forecasts Forecast bust probability - Probability of large forecast error over different regions Error-prone area detection - Identification of areas where …

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60. [SIH26080] Regime-Aware AI Post-Processing of Monsoon Rainfall Forecasts

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Disaster Management | **Submission Deadline:** 20 September 2026

• Problem Statement Rainfall forecast errors over India vary with weather regimes such as active monsoon, break monsoon, monsoon lows/depressions, orographic rainfall, coastal rainfall and western disturbances. A single bias-correction method may not work equally well in all situations.The challenge is to build an AI/ML-based rainfall post-processing system that first identifies the prevailing weather regime and then applies suitable correction to the raw NWP rainfall forecast.The aim is to improve district/grid-level rainfall forecasts, especially for heavy and very heavy rainfall events.
• Expected Outcome Expected Outcome - Description:
Weather regime classifier - Classification of active, break, depression,coastal/orographic rainfall regimes Bias-corrected rainfall forecast - Improved rainfall forecast compared to raw NWP output Heavy rainfall probability - Probability of rainfall …

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61. [SIH26081] Hybrid AINWP Multi-Model Forecast Blending System

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

• Problem Statement Different forecasting systems perform differently depending on region, season, lead time and weather situation. Physical NWP models, ensemble forecasts and AI/ML weather models may each have strengths under different conditions. Therefore, there is a need for an intelligent blending system that can dynamically combine multiple forecasts.
The challenge is to develop a hybrid AI–NWP blending framework that assigns adaptive weights to different forecast sources based on historical skill, forecast lead time, region, season and weather regime. The final product should provide an optimized forecast for rainfall, temperature, wind and extreme weather indicators.
Expected Outcome - Description
• Dynamically blended forecast - Best-combined forecast from multiple model sources
• Model weight maps - Indication of which model is more reliable for each region/lead time
• …

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62. [SIH26082] Air PollutionWeather Coupled Forecasting System (Delhi NCR Focus)

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Disaster Management | **Submission Deadline:** 20 September 2026

Traditional Air Quality Index (AQI) forecasting models typically treat meteorology and pollution dispersion as separate entities. However, in highly polluted urban landscapes like Delhi NCR, there is a critical, dynamic feedback loop between the weather and pollutants. During peak pollution seasons (such as the winter stubble-burning period), atmospheric inversion layers trap particulate matter close to the ground. Conversely, dense concentrations of aerosols (PM2.5) block sunlight,altering local temperatures, wind patterns, and planetary boundary layer (PBL) heights. Ignoring these coupled meteorological-chemical feedback loops leads to significant inaccuracies in standard AQI predictions. To achieve high-accuracy, actionable insights, there is an urgent need for an integrated system that simulates real-time interactions between atmospheric physics and chemical transport.
The challenge …

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63. [SIH26083] Extreme Heatwave Early Warning and Human Thermal Stress Index

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Disaster Management | **Submission Deadline:** 20 September 2026

In recent years, the frequency, duration, and intensity of heatwaves across India have escalated sharply due to climate change. However, traditional meteorological warnings rely almost exclusively on ambient dry-bulb temperature thresholds. This creates a critical vulnerability:standard forecasts ignore the deadly compounding effects of relative humidity, wind speed, and solar radiation on the human body.
A temperature of 40°C at 20% humidity feels vastly different from 40°C at 70% humidity-the latter can be fatal. Current public health infrastructure lacks localized, impact-based forecasting that translates raw weather data into actual physiological risk, human thermal stress levels, and projected mortality rates.
The challenge is to design an intelligent, localized early warning system that shifts heatwave forecasting from 'what the weather will be' to 'what the weather will do' to …

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64. [SIH26084] Convective scale nowcasting for Thunderstorms, Hail & Cloudbursts (06 hr)

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Disaster Management | **Submission Deadline:** 20 September 2026

Convective storms, such as severe thunderstorms, hail, downburst winds, and cloudbursts are among India’s deadliest natural hazards, especially during the pre-monsoon and monsoon seasons.Despite advancements in Numerical Weather Prediction (NWP) models, traditional systems often fail to accurately capture these mesoscale extreme weather events.The primary limitation stems from spatial and temporal constraints: these violent storms develop rapidly within a window of minutes and occur at localized scales that slip through coarse grid resolutions. Current early warning infrastructures struggle to provide high-resolution, short-term forecasts (0–6 hours), leaving local administrations, aviation sectors, and rural farming communities vulnerable to sudden, devastating impacts.
The challenge is to build a real-time, convective-scale Nowcasting System (0–6 hour lead time) operating at a …

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65. [SIH26085] Urban Flood Nowcasting System (Drainage and Rainfall Coupling)

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Disaster Management | **Submission Deadline:** 20 September 2026

Urban flooding in major Indian metros like Mumbai, Delhi, and Chennai has become an annual crisis. Traditional Numerical Weather Prediction (NWP) models fall short because knowing how much rain will fall does not automatically translate into knowing where the streets will flood.Urban flooding is a hyper-local phenomenon dictated by micro-topography, concrete imperviousness, and heavily strained, invisible drainage networks. Currently, municipal bodies lack real-time, street-level predictive systems. Consequently, cities are caught off guard by rapid water accumulation, leading to severe traffic gridlocks, economic disruption, and loss of life.
The challenge is to design a high-resolution, real-time Urban Flood Nowcasting System (0–3 hour lead time) capable of predicting street-level inundation before it happens.
Participants must move away from isolated weather models and instead build …

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66. [SIH26086] Hyperlocal Monsoon Onset & Break Prediction System (Block/Village Scale)

Organization: Ministry of Earth Sciences (MoES) | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

The Indian Summer Monsoon dictates the economic livelihood of millions of farmers, particularly during the Kharif sowing season. While macro-scale monsoon forecasts across large meteorological subdivisions have improved, Indian agriculture remains highly vulnerable to the unpredictable nature of intra-seasonal variations. Specifically, the exact dates of monsoon onset,prolonged dry spells (break-monsoon phases), and subsequent revival cycles vary drastically from one district to another.
Standard regional forecasts lack the spatial granularity required for localized agricultural planning.If a farmer sows seeds during a false onset just before a major breakthrough pause, entire crops fail due to moisture stress, leading to crushing financial losses.
The challenge is to build a hybrid predictive framework capable of delivering a 7-to-30-day probabilistic outlook of monsoon behavior at the …

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67. [SIH26089] Cooperative Gig Services Platform for Household & Community Services

Organization: Ministry of Cooperation | **Theme:** Smart Automation | **Submission Deadline:** 20 September 2026

• Background Labour Cooperative Federations and Labour Cooperative Societies possess a large pool of skilled workers such as electricians, plumbers, carpenters, painters, domestic helpers, caregivers, drivers, gardeners,cleaners, and technicians. However, they lack a structured digital platform to connect these workers with households and institutions requiring such services.Private platforms currently dominate this market, while cooperative workers often remain underutilized despite having skills and local presence.
• Problem Statement To develop a cooperative-owned digital service marketplace platform that enables Labour Cooperative Federations and Labour Cooperative Societies to provide verified household and community services while ensuring fair wages, worker welfare, and consumer trust.
• Expected Solution Features
• Service provider registration and verification
• Worker skill …

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68. [SIH26090] AI-Driven Market Linkage and Smart Cataloging Mobile Application for Marginalized Artisans

Organization: Ministry of Social Justice and Empowerment (MoSJE) | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

• Background The government actively supports the socio-economic upliftment of marginalized communities,particularly micro-entrepreneurs, artisans, and weavers. Financial assistance is provided to establish small-scale manufacturing and handicraft units. To help these beneficiaries sell their goods, market exposure is facilitated through periodic physical exhibitions, cluster development programs, and trade fairs (such as Shilp Samagam, Surajkund Mela, and Dilli Haat).While physical exhibitions provide a temporary boost in sales, these micro-entrepreneurs lack continuous, year-round access to broader digital markets. Transitioning to the digital economy is hindered by low digital literacy, language barriers, and a lack of technical skills required to professionally photograph, price, and catalog products for modern e-commerce.
• Challenge There is a critical need to bridge the gap …

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69. [SIH26091] AI-Driven Hyper-Local Business Advisory and Financial Structuring Assistant for Rural Micro-Entrepreneurs

Organization: Ministry of Social Justice and Empowerment (MoSJE) | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

• Background The government actively promotes the economic empowerment of marginalized communities by providing concessional credit for income-generating activities. Under various schemes,beneficiaries are required to contribute a small margin money fraction-typically 10% of the total project cost-while the State Channelizing agencies (SCAs) Channelizing agencies(CAs) provide the remaining 90% as a concessional loan. For example, if an entrepreneur wishes to establish a ?10,00,00 enterprise, they must possess ?1,00,00 as their 10% contribution, making them eligible for a ?9,00,00 loan.
These loans are categorized into specific tiers:
? Micro Finance Scheme: For small units with a project cost up to ?1.40 lakh. The funding agency provides up to 90% (maximum ?1.25 lakh) at a concessional interest rate of 6.5% per annum for the beneficiary, to be repaid over 3 years (including a 3-month …

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70. [SIH26092] AI-Driven Scheme Matching for Marginalized Entrepreneurs

Organization: Ministry of Social Justice and Empowerment (MoSJE) | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

• Background To promote the socio-economic empowerment of the Scheduled Caste (SC) population, the government provides concessional financial assistance and educational loans. Beneficiaries with an annual family income of up to ?5.00 Lakhs are eligible for various tailored financial products covering up to 90% of their project or education costs at highly concessional interest rates (typically 6.5% to 8% per annum).
However, direct loan applications are not entertained. Instead, funds are routed through a 'Channel Finance System' comprising over 100 Channel Partners, including State Channelizing Agencies (SCAs), Public Sector Banks (PSBs), Regional Rural Banks (RRBs), and NBFC-MFIs.
• Challenge Citizens often lack awareness regarding which specific credit scheme fits their needs-such as distinguishing between a Micro Finance Scheme for small projects (up to ?1.40 lakh), a Term Loan for …

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71. [SIH26093] AI-Based Real-Time Stress and Trauma Assessment Module for Victims/Complainants Accessing NHAA (14566) and Integrated Portal

Organization: Ministry of Social Justice and Empowerment (MoSJE) | **Theme:** Smart Automation | **Submission Deadline:** 20 September 2026

• Background Victims and complainants belonging to Scheduled Castes and Scheduled Tribes who approach the National Helpline Against Atrocities (14566), Integrated Portal, chatbot, mobile application, IVRS, or other digital platforms often experience severe emotional distress arising from caste-based discrimination, violence, rape, gang rape, murder of family members, social boycott, displacement, threats, and prolonged legal proceedings. Presently, there is no standardized mechanism for assessing the psychological condition and vulnerability of victims at the time of first contact with authorities.
• Problem Statement Design and develop an AI-enabled Real-Time Stress and Trauma Assessment Module that can assess the psychological stress, trauma, fear, anxiety, and vulnerability levels of victims/complainants interacting through NHAA (14566), the Integrated Portal, chatbot,IVRS, mobile …

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72. [SIH26094] AI-Powered Dynamic Mental Health Monitoring and Distress Prediction System for Victims of Atrocities

Organization: Ministry of Social Justice and Empowerment (MoSJE) | **Theme:** MedTech / BioTech / HealthTech | **Submission Deadline:** 20 September 2026

• Background Victims of atrocities frequently experience prolonged psychological distress after complaint registration due to threats, intimidation, repeated court appearances, delays in investigation and trial, social ostracism, economic hardship, and rehabilitation challenges.Existing mechanisms focus primarily on legal and financial support and do not provide continuous monitoring of victim well-being.
• Problem Statement Develop an AI-based Dynamic Mental Health Monitoring and Distress Prediction System that continuously monitors and predicts psychological distress among victims and complainants registered through NHAA (14566), the Integrated Portal, chatbot, mobile application, IVRS, or other approved communication channels throughout the investigation,trial, rehabilitation, and compensation process.
• Expected Solution The system should:
• Conduct periodic interactions with …

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73. [SIH26095] Smart Real-Time Monitoring & Inspection Mobile App

Organization: Ministry of Social Justice and Empowerment (MoSJE) | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

• Problem Statement Develop a centralized mobile application for real-time monitoring, surprise inspections, CCTV surveillance integration, and random inspection assignment for projects/institutes/NGOs running under DoSJE schemes.
• Key Features
• Live CCTV feed integration from projects/institutes
• Random Video Conferencing (VC) connectivity with Project Incharge/Staff/Beneficiaries
• Real-time monitoring dashboard for Department officials
• Mobile-based inspection module for PMU/Inspection Teams
• Random assignment of inspection duties through AI/automation
• Geo-tagged inspection reports and live evidence capture
• AI-based anomaly and attendance analytics
• Stakeholders
• DoSJE Divisions
• PMU Teams
• NGOs/Institutes
• Beneficiaries
• State/District Authorities
• Expected Outcomes:
• Improved transparency and accountability
• Reduction in fake reporting and proxy functioning
• …

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74. [SIH26097] AI-Driven voice Assistant for livelihood Mapping and NSQF-Aligned Skilling Recommendations for SC Communities under GIA component of PM-AJAY

Organization: Ministry of Social Justice and Empowerment (MoSJE) | **Theme:** Smart Education | **Submission Deadline:** 20 September 2026

• Background
• The Pradhan Mantri Anusuchit Jaati Abhyuday Yojana (PM-AJAY) aims to reduce poverty among Scheduled Caste (SC) communities through livelihood promotion, skill development, and enterprise support under its Grant-in-Aid (GIA) component. A major challenge in implementation is the identification of appropriate skill training pathways that align with both the aspirations of beneficiaries and the actual livelihood opportunities available in their local regions.
• Many target beneficiaries face barriers such as low digital literacy, limited awareness of modern trades, language constraints, and difficulty navigating text-heavy digital systems. As a result, there is often a mismatch between enrolled training programs and the beneficiary’s interests, capabilities, or local market demand, leading to high dropout rates and poor post-training employment outcomes.
• To improve …

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75. [SIH26099] AI-Driven Standardization and Harmonization of Material Codes Across CPSEs

Organization: Ministry of Petroleum & Natural Gas | **Theme:** Smart Automation | **Submission Deadline:** 20 September 2026

• Background Central Public Sector Enterprises (CPSEs) operating in sectors such as Oil & Gas, Power, Steel, Mining and Heavy Engineering procure and maintain a large number of similar or functionally equivalent materials. However, the same material may be assigned different material codes, descriptions, specifications, units of measurement and classification across different CPSEs.
This results in duplication of material masters, inconsistent descriptions, difficulty in identifying equivalent materials, fragmented procurement data, higher inventory levels and limited opportunities for collaborative procurement.
A unified and intelligent approach is therefore required to standardize, harmonize and rationalize material master data across CPSEs.
• Description The proposed solution envisages development of an AI-powered National Unified Material Master Framework capable of analysing …

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76. [SIH26100] AI-Powered Integrated Bid Compliance Verification Platform for GeM Procurement

Organization: Ministry of Petroleum & Natural Gas | **Theme:** Smart Automation | **Submission Deadline:** 20 September 2026

• Background Government procurement through the Government e-Marketplace (GeM) involves verification of multiple statutory, regulatory and eligibility requirements of bidders.
Procurement officers are required to examine and validate documents and information related to Udyam/MSME registration, GST registration and return filing, PAN and Income Tax compliance, Make in India/local content, EPFO/ESIC compliance, Startup India, NSIC, OEM authorization, DigiLocker, blacklisting/debarment and other applicable statutory requirements.
The verification process is largely document-intensive and requires cross-checking information across multiple government portals and databases. This results in significant manual effort, longer tender evaluation time and the possibility of inconsistencies or human errors.
• Description The problem statement envisages development of an AI-powered integrated bid …

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77. [SIH26101] Develop an AI enabled learning platform that identifies competency gaps, recommends personalized training through integration with the iGOT Karmayogi ecosystem, and capable of generating Quizzes and Multiple choice questions (MCQs) from uploaded learning materials to strengthen capacity building in India's Official Statistical System.

Organization: MoSPI | **Theme:** Smart Education | **Submission Deadline:** 20 September 2026

• Background India's statistical system is undergoing rapid technology advancement with increasing adoption of Artificial Intelligence (AI), Machine Learning (ML), Big Data Analytics, GIS, cloud computing, and modern statistical methodologies. Officials engaged in data collection, processing, analysis, dissemination, and policy support require continuous upskilling to meet evolving technological and domain-specific requirements.
While the iGOT Karmayogi platform offers a vast repository of learning resources, officials often face challenges in identifying the most relevant courses aligned with their job roles, current competencies, and future skill requirements.
Presently, there is no intelligent mechanism that performs comprehensive skill-gap assessment and recommends personalized learning pathways specifically for professionals working in Official Statistics.
Artificial Intelligence …

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78. [SIH26102] Development of an AI-powered system to detect anomalies, fraud, and inefficiencies in MPLAD Scheme implementation regd.

Organization: MoSPI | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

• Background The Members of Parliament Local Area Development Scheme (MPLADS) is a Central Sector Scheme under which Hon'ble Members of Parliament recommend developmental works for creation of durable community assets and provision of basic civic amenities. The Scheme involves large-scale fund utilization and execution of thousands of works across the country through multiple implementing agencies and administrative authorities. Given the volume and complexity of financial and project-related data generated under the Scheme, there is a need for an AI-powered solution that can leverage machine learning and advanced analytics to detect trends and anomalies in expenditure patterns, fund utilization, cost estimates, and work execution, thereby enabling early identification of potential fraud, inefficiencies, and non-compliance while enhancing transparency, accountability, and effective …

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79. [SIH26103] Use case on web-based integrated project-monitoring platform

Organization: MoSPI | **Theme:** Smart Automation | **Submission Deadline:** 20 September 2026

• Background The Infrastructure & Project Monitoring Division (IPMD), Ministry of Statistics and Programme Implementation (MoSPI) monitors the Central Sector Infrastructure Projects costing ?150 crore and above, across all the infrastructural Ministries/ Departments. The project monitoring was undertaken through the Online Computerised Monitoring System (OCMS) since 2006, which served as the primary repository of project-level information relating to project cost, expenditure, timelines and implementation status. Over nearly two decades, OCMS generated a valuable historical database capturing project implementation trends, cost overruns and time overruns across sectors. Later, OCMS was modernized to Project Assessment, Infrastructure Monitoring and Analytics for Nation-building (PAIMANA) portal, to enable a comprehensive and integrated project-monitoring ecosystem.
• PAIMANA Portal and …

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80. [SIH26104] AI-Powered Real-Time Detection and Prevention of Voice Cloning Impersonation Attacks

Organization: All India Council for Technical Education (AICTE) | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

• Background Recent advancements in generative AI and neural speech synthesis have made high-fidelity voice cloning possible from only a few seconds of recorded audio. Threat actors are exploiting these capabilities to impersonate CXOs,government officials, and trusted individuals in order to initiate fraudulent financial transactions, manipulate employees,or bypass verification procedures in high-risk workflows. Conventional call verification methods-such as caller ID, manual call-back, and basic voice familiarity-are no longer sufficient to distinguish genuine callers from AI-generated or manipulated voices, especially in high-pressure social engineering scenarios.
These attacks are increasingly orchestrated over VoIP, mobile networks, and enterprise collaboration platforms,sometimes combined with leaked personal information to create highly convincing narratives. The absence of …

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81. [SIH26105] AI-Powered Continuous Cyber Risk Quantification and Investment Optimization Platform

Organization: All India Council for Technical Education (AICTE) | **Theme:** Blockchain & Cybersecurity | **Submission Deadline:** 20 September 2026

• Background Enterprises and institutions invest heavily in cybersecurity tools, compliance programs, and risk management initiatives, yet cyber risk is still predominantly communicated using qualitative ratings such as 'Low','Medium', or 'High'.
These coarse categories fail to express the potential financial impact of cyber threats,making it difficult for senior management, boards, and regulators to evaluate whether current cyber investments are adequate or optimally allocated. Cyber risk is inherently dynamic: new vulnerabilities emerge, threat actors change tactics, business services are added or retired, and security controls mature over time. Most current risk assessment practices rely on periodic, manual exercises, resulting in stale risk registers and limited visibility into the organization's real-time cyber exposure. This gap leads to suboptimal prioritization of remediation …

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82. [SIH26106] AI-Powered Email Threat Detection, GeoLocation and Forensic Intelligence Platform

Organization: All India Council for Technical Education (AICTE) | **Theme:** Blockchain & Cybersecurity | **Submission Deadline:** 20 September 2026

• Background Email continues to be one of the most widely used communication channels in government, education, banking,and enterprise ecosystems. However, it also remains one of the most exploited attack vectors for phishing,impersonation, business email compromise, financial fraud, credential theft, and malware delivery. Threat actors increasingly use spoofed domains, deceptive sender identities, social engineering techniques, and compromised infrastructure to send highly convincing fraudulent emails that appear legitimate to end users.Traditional email security controls such as spam filters, static blacklists, and rule-based signature mechanisms are often insufficient to detect sophisticated fraudulent emails. Attackers now use AI-generated language,domain lookalikes, display-name spoofing, hidden redirection links, and relay chains to evade standard detection systems. In many cases, …

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83. [SIH26107] AI-powered Intelligent Assistant for Indian Standards and BIS Services for Industries and Consumers

Organization: Ministry of Consumer Affairs, Food & Public Distribution | **Theme:** Smart Automation | **Submission Deadline:** 20 September 2026

• Background The Bureau of Indian Standards publishes thousands of Indian Standards and provides various services such as product certification, hallmarking, laboratory recognition, Standards Clubs, training, consumer affairs, and conformity assessment.
• At present, users often struggle to identify:
• Applicable Indian Standards for their products,
• Certification requirements,
• Relevant BIS schemes,
• Licensing procedures,
• Testing requirements,
• Related standards, and
• Answers to technical queries.
Searching through multiple documents, portals, and PDFs is time-consuming, particularly MSMEs, startups, students, and consumers.
• Description Develop an AI-powered conversational assistant that enables users to obtain accurate, context-aware, and source-backed information related to Indian Standards and BIS services through natural language interactions.
The assistant should …

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84. [SIH26108] AI-Powered Recommendation Engine for Identifying Applicable Indian Standards for Procurement Specifications

Organization: Ministry of Consumer Affairs, Food & Public Distribution | **Theme:** Smart Automation | **Submission Deadline:** 20 September 2026

• Background Government departments, Public Sector Enterprises (PSES),procurement agencies, and private organizations procure a wide range of products and services through e-procurement portals. Procurement officials are often required to prepare technical specifications that reference the appropriate Indian Standards (IS).However, identifying the correct standard(s) is challenging due to the large number of published standards, overlapping scopes, frequent revisions, and the need to consider associated or normative reference standards. Consequently, tender specifications may omit relevant standards, reference outdated versions, or include incomplete technical requirements, leading to ambiguity, reduced product quality, and procurement disputes.
An intelligent system is required that can automatically analyze a product description or technical specification and recommend the most …

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85. [SIH26111] Smart AI-Enabled Rapid Feed and Silage Quality Testing System for Dairy Farmers

Organization: Ministry of Fisheries, Animal Husbandry & Dairying | **Theme:** Agriculture, FoodTech & Rural Development | **Submission Deadline:** 20 September 2026

• Background Animal nutrition directly affects milk production, animal health, reproductive performance, and dairy profitability. Dairy farmers often face challenges due to poor-quality cattle feed,adulterated feed ingredients, fungal contamination, toxin presence, and low-quality silage.Conventional feed testing laboratories are expensive and inaccessible for many rural farmers.There is a need for rapid, portable, affordable, and digitally enabled feed quality assessment systems.Emerging technologies such as AI, IoT, spectroscopy, computer vision, and biosensors can help create real-time feed testing and advisory systems for dairy farmers.
• Description Participants are required to develop a rapid digital testing solution capable of:
• Assessing nutritional quality of cattle feed and silage;
• Detecting adulteration and contamination;
• Providing instant farmer advisories and feed …

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86. [SIH26114] Smart City Site Planning using Autodesk Forma Site Design

Organization: Autodesk | **Theme:** Smart Automation | **Submission Deadline:** 20 September 2026

• Description
• Students are tasked with designing a Smart City using Forma Site Design for effective site design for adding details to the buildings in the site.
• The Site area selected must have minimum 1 km square area
• The goal is to design and develop a site including the contextual data (available for free within Forma Site Design) and export Site BIM model suitable for further use in Revit app.
• The Site designed must include Site Limits, Landscaping, Buildings, and Transportation elements.
• The Site designed must be analyzed using the Analyze functions available within Forma Site Design for Area Metrics, Embodied Carbon, Sun hours, Daylight potential, Wind Analysis, Microclimate analysis, Noise analysis, Solar Energy.
• Minimum 2 Site Design proposals must be compared and presented using the Forma Borad available within Forma Site Design.
• Objective Urban populations …

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87. [SIH26115] Design and Develop a Smart Mobile Medical-Waste Collection and Segregation System

Organization: Autodesk | **Theme:** MedTech / BioTech / HealthTech | **Submission Deadline:** 20 September 2026

• Description Healthcare facilities generate large volumes of biomedical waste that require safe, compliant, and efficient handling. Manual collection and segregation increase the risk of contamination, operational inefficiencies, and regulatory challenges.
Design and develop an AI-powered, battery-electric autonomous mobile system that automates the collection, identification, segregation, and digital tracking of biomedical waste across hospitals. The solution should leverage AI-enabled vision systems to classify waste, intelligently segregate it into designated compartments, and provide end-to-end traceability while minimizing human exposure to hazardous materials and improving safety, operational efficiency, and regulatory compliance.
Using Autodesk Fusion, students must demonstrate a complete product development lifecycle-from concept ideation to manufacturing-ready product, …

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88. [SIH26116] Urban Mixed-Use Design Challenge-Design a centrally located mixed-use building in Autodesk Revit with commercial spaces (Ground + 1st floor) and residential units (up to 8 floors). 1 Level of Basement (Car Parking + EV Charging), Total (B+G+9)(Note: Plot size and all required dimensions may be assumed by students (in mm units).

Organization: Autodesk | **Theme:** Smart Education | **Submission Deadline:** 20 September 2026

• Description
• Facade-Driven Architectural Expression-Develop an innovative facade system that enhances architectural aesthetics, responds to climate (light, heat, ventilation), and blends with the surrounding urban context.
• Breathable & Nature-Integrated Design-Incorporate a central landscape courtyard and green interfaces (terraces, balconies) to create an airy, breathable structure that integrates nature and improves occupant well-being.
• Residential & Commercial Design Efficiency-Ensure functional planning for commercial activation on lower floors and well- designed residential units above, with optimal daylight, ventilation, privacy, and views.
• Structural Modeling & Detailing-Create 2D structural drawings for key components such as beams, columns, and slabs, including necessary detailing.The model should include all essential building elements: beams, columns, slabs, stairs, …

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89. [SIH26117] Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work

Organization: Mangalore Refinery and Petrochemicals Limited (MRPL) | **Theme:** Smart Automation | **Submission Deadline:** 20 September 2026

• Background Refineries, PSUs, defence-linked manufacturing units and government offices generate a lot of routine but sensitive knowledge work. Approval notes, board presentations, engineering calculations, code for internal tools, review of scanned drawings and inspection reports. None of this can go through cloud AI assistants like Claude or Codex because the underlying data is confidential: Piping & Instrument Diagrams, financials, vendor negotiations, unreleased designs, internal correspondence, confidential business strategies etc. Company policy keeps this data on premises, so people either do the work manually resulting in productivity gain, or they quietly paste confidential material into public tools anyway. Open weight large reasoning models have reached a point where a genuinely useful assistant built on them is realistic. But nothing deployable exists today that industrial …

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90. [SIH26119] Indigenous GPU-Accelerated Optimization Solver (Sovereign Alternative to Express / CEPLEX)

Organization: Mangalore Refinery and Petrochemicals Limited (MRPL) | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

• Background Almost every optimization problem in India's refining, petrochemical, power, logistics, manufacturing and planning sectors ultimately depends on a handful of foreign mathematical optimization solvers such as IBM ILOG CPLEX, Gurobi and FICO Xpress. These engines sit behind refinery scheduling, production planning, supply chain optimization, blending, energy management and many AI-driven decision-support systems. While they are extremely capable, they come with high recurring license costs, restrictive licensing models and limited visibility into the underlying optimization algorithms. Indian developers can formulate optimization problems, but they cannot inspect, modify or tailor the solver internals to suit strategic national requirements. Open-source alternatives such as COIN-OR CBC, HiGHS, GLPK and SCIP exist and have made significant progress, but they still lag behind …

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91. [SIH26120] Digital Twin for Well-to-Surface Optimization of Cyclic Steam Stimulation (CSS) and Sucker Rod Pump (SRP) Operations for Heavy Oil Wells of Baghewala Field.

Organization: Oil India Limited | **Theme:** Smart Automation | **Submission Deadline:** 20 September 2026

• Background Baghewala Field in Rajasthan produces heavy crude oil (17–19° API) from the Jodhpur Sandstone reservoir. The reservoir is characterized by High crude viscosity, High asphaltene content, Low reservoir pressure, Low reservoir temperature (46–48°C) and Poor oil mobility under primary recovery. Consequently, artificial lift and thermal enhanced oil recovery are critical for sustained production. At present, CSS cycle design and SRP operation are optimized separately using historical experience. As reservoir temperature declines after steam injection, crude viscosity increases, leading to reduced pump efficiency, higher energy consumption, rod floating issues, rod failures and lower oil recovery. There is a need for an integrated, data-driven system that continuously optimizes both CSS and artificial lift operations.
• Problem Description Current operations face the following …

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92. [SIH26121] eRTMAC-NWIS (Nearby Wells Intelligence System): An AI-Powered Offset Well Knowledge and Decision Support Platform for Drilling Operations

Organization: Oil India Limited | **Theme:** Smart Automation | **Submission Deadline:** 20 September 2026

• Background Oil India Limited has a digital real-time monitoring system (eRTMAC) that provides real-time drilling data, mud logging information, and wellsite analytics across operational areas. However, drilling decisions, particularly in geologically complex formations, require not only real-time data from the active well but also insights from nearby and historical wells drilled in the same reservoir or formation. Historical drilling knowledge currently resides across numerous well completion reports, drilling reports, PDF documents, and individual experience, making retrieval time-consuming and dependent on individual experience &
memory. This often results in delays in decision-making and missed opportunities to proactively mitigate drilling risks.
• Problem Description Currently, drilling teams do not have a unified platform that can:
i. Display nearby wells on a geospatial map …

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93. [SIH26122] Intelligent Data Capture & Schedule-Linking Layer for Infrastructure Project Management: Real-Time Actual Progress Tracking (Planning-to-Execution Bridge)

Organization: Oil India Limited | **Theme:** Smart Automation | **Submission Deadline:** 20 September 2026

• Background Infrastructure project schedules cascade from macro milestones (L1) down to micro, executable activities (L5/L6),spanning multiple engineering disciplines - civil, piping, static/rotating equipment, electrical, instrumentation, HSE- each executing and reporting in parallel. While the baseline plan is well-structured (Primavera/MS Project), actual execution data flows back through daily progress reports, site diaries, discipline-wise spreadsheets, and verbal supervisor updates, each in its own format and cadence, largely disconnected from the L5/L6 activity IDs in the plan.
• Problem Description There is no reliable, low-friction mechanism to capture actual start/end times of L5/L6 activities across disciplines and auto-link them back to the plan. Input quality varies with manpower skill, reporting discipline, and format.Field execution is often more granular than the …

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94. [SIH26123] Edge-AI Based Distributed Fleet Coordination for Autonomous Mobile Robots (AMRs) in Smart Warehouses

Organization: Bharat Electronics Limited | **Theme:** Robotics and Drones | **Submission Deadline:** 20 September 2026

• Background Modern smart warehouses rely on fleets of Autonomous Mobile Robots (AMRs) to move goods efficiently. As fleet sizes grow, relying entirely on a centralized cloud server for path planning causes high network latency, Wi-Fi dead-zone vulnerabilities, and single-point-of-failure risks.To ensure continuous operation, modern robotics is shifting toward decentralized, edge-computing solutions where robots can talk to each other directly and make split-second decisions on the fly.
• Description The objective is to design a decentralized coordination and collision-avoidance framework for a multi-robot fleet (at least 3 AMRs) operating in a dynamic warehouse environment. The system must run locally on edge hardware (e.g., Raspberry Pi or Jetson Nano onboard each robot) and handle:
1. Decentralized Communication: Inter-robot messaging to share position and intent without a central …

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95. [SIH26124] AI-Powered Mobile Urban Intelligence Platform Using Public Transport Fleet

Organization: Bharat Electronics Limited | **Theme:** Fitness & Sports | **Submission Deadline:** 20 September 2026

• Background Urban public transport buses traverse almost every major road in a city every day. Modern buses are increasingly equipped with multiple cameras covering the front, rear, sides, and passenger cabin. However, these cameras are primarily used for recording incidents and are not leveraged as intelligent sensing platforms. At the same time, city authorities rely on fixed CCTV cameras, manual inspections and citizen complaints to identify road defects, traffic congestion,missing infrastructure and unsafe driving behaviour. This results in delayed response,incomplete situational awareness and inefficient maintenance planning.
• Description Develop an AI-powered onboard and centralized software platform that transforms public transport buses into mobile urban sensing units. The onboard software shall analyse video streams from multiple bus-mounted cameras to detect road defects …

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96. [SIH26125] Blockchain-Based Secure Platform for Identity,Access Control, and Digital Asset Management

Organization: Bharat Electronics Limited | **Theme:** Blockchain & Cybersecurity | **Submission Deadline:** 20 September 2026

• Background Organizations today rely heavily on centralized identity and access management systems, which create significant security and operational risks. These systems are vulnerable to cyber attacks, identity theft, unauthorized access, and single points of failure. Additionally, digital and physical asset ownership is often managed through disconnected or semi-centralized systems, making verification of authenticity, access rights, and ownership history difficult and unreliable. There is a growing need for a decentralized, tamper-proof system that can securely manage user identities, control access permissions, and ensure transparent ownership of digital assets
• Detailed Description The system aims to introduce a blockchain-based framework that integrates decentralized identity management, access control, and NFT-based digital asset ownership. Each user is assigned a …

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97. [SIH26126] Vision Based Autonomous Navigation for Unmanned Ground Vehicle for Outdoor environment

Organization: Bharat Electronics Limited | **Theme:** Robotics and Drones | **Submission Deadline:** 20 September 2026

• Background Outdoor Unmanned Ground Vehicles (UGVs) face unpredictable terrain, changing light, and unreliable GPS signals. To achieve true autonomy in applications like search-and-rescue,agriculture, or delivery, UGVs must rely on onboard computer vision. Visual perception provides a cost-effective, data-rich way for vehicles to understand and safely navigate complex,unstructured outdoor surroundings.
• Description The objective is to build an autonomous navigation system for a UGV operating in a GPS-denied outdoor environment using camera feeds as the primary sensor. Students must solve three key challenges:
1. Path Detection: Real-time identification of safe, traversable paths vs. hazards (e.g., rocks,ditches, trees).
2. Visual Localization: Estimating the UGV’s position and orientation without GPS using visual data.
3. Collision Avoidance: Dynamically routing the vehicle around …

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98. [SIH26127] City-Wide AI Engine for Multi-Camera ANPR Trajectory Tracking and Urban Traffic Analytics

Organization: Bharat Electronics Limited | **Theme:** Transportation & Logistics | **Submission Deadline:** 20 September 2026

• Background Modern urban centers deploy vast networks of CCTV and Automatic Number Plate Recognition(ANPR) cameras to manage traffic, enforce traffic laws, and maintain public security. However,most existing systems process these feeds in isolated silos, performing basic license plate detection without effectively linking data across space and time. This lack of integration prevents city authorities from automatically tracking high-interest vehicles across different sectors and limits their ability to extract macro-level traffic movement trends from the existing camera infrastructure.
• Description The objective is to develop a robust, centralized AI software platform that processes multicamera feeds across a city-wide ANPR network to accomplish three core functionalities. First, the platform must feature a High-Accuracy ANPR and OCR Engine, which utilizes an advanced Optical Character …

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99. [SIH26128] Efficient systems for early detection,prevention,and management of livestock diseases and animal health issues

Organization: Government Of Maharashtra | **Theme:** Agriculture, FoodTech & Rural Development | **Submission Deadline:** 20 September 2026

• Problem Description Livestock owners, field veterinarians,para-veterinary workers and government departments often lack a unified, realtime mechanism to identify emerging animal-health risks at the village, block and district levels. Disease symptoms may be reported late, diagnostic facilities may be distant, vaccination and treatment histories may be incomplete, and information from farms, veterinary dispensaries, laboratories, vaccination drives and surveillance programmes may remain fragmented. These gaps can delay containment, increase livestock mortality and productivity loss, raise the risk of zoonotic transmission, and affect farmers' incomes. The challenge is to create a practical system that enables early warning, rapid reporting, risk assessment, preventive action, referral and coordinated response, including in low-connectivity areas.
• Expected Solution / Outcome A …

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100. [SIH26129] System integration and interoperability among government digital platforms,resulting in fragmented service delivery

Organization: Government Of Maharashtra | **Theme:** Smart Automation | **Submission Deadline:** 20 September 2026

• Problem Description Government departments operate multiple portals, mobile applications,registries, workflow systems and databases that have often been developed independently. Differences in data formats, identifiers, authentication methods, APIs, process definitions and ownership structures can prevent seamless information exchange.Citizens and businesses may be required to submit the same information repeatedly, track applications across different portals, or visit multiple offices.Officials may lack a consolidated view of beneficiaries, applications, approvals,grievances and service outcomes. The challenge is to enable secure,standards-based interoperability without requiring complete replacement of existing systems.
• Expected Solution / Outcome An interoperability framework,middleware layer or federated service delivery architecture that supports API based exchange, common data …

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101. [SIH26130] Efficiency in streamlining industrial approvals,compliance processes,and access to government support services

Organization: Government Of Maharashtra | **Theme:** Smart Automation | **Submission Deadline:** 20 September 2026

• Problem Description Entrepreneurs and industrial units may need to obtain multiple registrations,permissions, licences, no-objection certificates, inspections and renewals from different authorities. Requirements may vary by sector, location, project size and stage of operation. Applicants may find it difficult to identify applicable approvals, understand documentation requirements, monitor timelines,respond to queries and access incentives or support schemes. Departments may face incomplete applications, repetitive scrutiny, manual coordination, limited visibility of bottlenecks and inconsistent compliance monitoring. The challenge is to simplify and accelerate the end-to-end journey while maintaining statutory safeguards.
• Expected Solution / Outcome A unified, intelligent approval and compliance management solution that can generate a customised approval checklist, guide …

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102. [SIH26131] Early detection and management of crop diseases and pest infestations

Organization: Government Of Maharashtra | **Theme:** Agriculture, FoodTech & Rural Development | **Submission Deadline:** 20 September 2026

• Problem Description Farmers often recognise crop diseases or pest infestations only after visible damage has spread. Extension staff may cover large areas, while laboratory diagnosis and expert advice may not be immediately available. Weather, crop stage, variety, soil condition and local pest history influence risk, but these inputs are rarely combined into actionable farm-level alerts. Incorrect diagnosis may lead to delayed treatment, excessive or inappropriate pesticide use, increased cultivation cost,residue concerns and yield loss. The challenge is to provide timely, reliable and locally relevant detection,forecasting and management support.
• Expected Solution / Outcome A farmer- and extension-worker-friendly crop-health system that supports image based symptom identification, pest-trap or sensor inputs, weather-based risk forecasting, geospatial hotspot mapping,expert …

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103. [SIH26132] Strengthening market linkages and price discovery for farmers

Organization: Government Of Maharashtra | **Theme:** Agriculture, FoodTech & Rural Development | **Submission Deadline:** 20 September 2026

• Problem Description Many farmers, especially smallholders and producer groups, have limited visibility of current and expected prices across nearby markets, processors,institutional buyers and digital trading channels. Information on quality specifications, demand, logistics,storage, payment reliability and buyer credentials may be fragmented. Farmers may sell immediately after harvest because of liquidity or storage constraints and may have weak bargaining power. Buyers, meanwhile,may struggle to aggregate consistent volumes and verify quality. The challenge is to improve transparent price discovery and create reliable, efficient linkages from farm gate to suitable buyers.
• Expected Solution / Outcome A market-intelligence and transaction enablement solution that aggregates mandi prices, buyer demand, quality requirements, arrival volumes, transport and storage options; provides …

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104. [SIH26133] Accessibility and quality of public healthcare services,particularly in rural and underserved areas

Organization: Government Of Maharashtra | **Theme:** MedTech / BioTech / HealthTech | **Submission Deadline:** 20 September 2026

• Problem Description Rural and underserved communities may face long travel distances,shortages of specialists, irregular diagnostics, fragmented medical records, delayed referrals and limited awareness of available services. Primary health facilities may have constrained staff and equipment, while patients may move between sub-centres, primary health centres, rural hospitals and district hospitals without continuity of information. Connectivity, language,health literacy and affordability further affect access. The challenge is to improve timely access, continuity, quality and accountability while strengthening-not replacing-the public-health system.
• Expected Solution / Outcome An integrated care-access and quality support solution that may combine assisted teleconsultation, appointment and queue management, digital triage,longitudinal patient records, referral tracking, diagnostic …

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105. [SIH26134] Challenges in aligning skill development programs with industry requirements and emerging job market demands

Organization: Government Of Maharashtra | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

• Problem Description Skill-development programmes may be designed using broad or historical occupation categories that do not fully reflect changing technologies, local industry demand, job roles, productivity standards and employer expectations.Course curricula, equipment, trainer capacity and assessment methods may lag emerging requirements. Employers may struggle to identify job-ready candidates, while trainees may complete courses that have limited placement potential. The challenge is to create a continuous, evidence-based mechanism for translating industry demand into course design, capacity planning, trainer development and candidate guidance.
• Expected Solution / Outcome A labour-market intelligence and curriculum-alignment platform that combines job-posting signals, employer surveys, industry consultations, sector growth data, placement outcomes and emerging-technology trends …

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106. [SIH26135] Difficulties in tracking employment outcomes,skill gaps, and the impact of skilling initiatives

Organization: Government Of Maharashtra | **Theme:** Smart Education | **Submission Deadline:** 20 September 2026

• Problem Description Training systems frequently capture enrolment, attendance, assessment and certification, but reliable information on employment, self-employment, job retention, wage progression, relevance of training and longer-term livelihood outcomes may remain incomplete.Trainees may change phone numbers or locations, employers may not report consistently, and multiple programmes may use different identifiers and definitions. Without longitudinal outcomes, it is difficult to compare providers, improve courses, target future investments or demonstrate public value. The challenge is to establish credible, low-burden and privacy conscious outcome tracking.
• Expected Solution / Outcome A longitudinal skilling-outcomes and impact-measurement system that creates consent-based trainee records,links training with placement and employment signals, conducts automated and assisted …

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107. [SIH26136] Startup friendly public procurement mechanism that enables government departments to identify,pilot, procure,and scale innovative solutions from eligible startups

Organization: Government Of Maharashtra | **Theme:** Smart Automation | **Submission Deadline:** 20 September 2026

• Problem Description Government departments often face operational problems that could benefit from innovative startup solutions, but conventional procurement processes are generally designed for standardised goods and established vendors.
Departments may find it difficult to formulate outcome-based problem statements, discover suitable startups, evaluate novel technologies, structure controlled pilots, manage intellectual property and data, measure pilot results, and transition successful pilots into compliant procurement or scale-up.Startups may struggle with prior-turnover or experience requirements, long sales cycles, unclear payment milestones and limited visibility of departmental demand. The challenge is to create a transparent, competitive and legally compliant innovation-procurement pathway.
• Expected Solution / Outcome A structured end-to-end mechanism for challenge …

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108. [SIH26137] Quantum-Inspired Intelligent Traffic Route Optimization in Transportation Systems Using Metaheuristic Optimization

Organization: Egreen Quanta | **Theme:** Fitness & Sports | **Submission Deadline:** 20 September 2026

Background Modern urban transportation networks face persistent challenges of traffic congestion, inefficient route planning, and high operational costs. Classical optimization techniques struggle with large-scale Vehicle Routing Problems (VRP) because of their NP-hard nature. While quantum computers offer theoretical advantages for combinatorial optimization,current hardware limitations prevent their direct large-scale use. Quantum-inspired metaheuristic algorithms (e.g., Quantum Particle Swarm Optimization - QPSO) embed quantum-mechanical concepts into classical computation, delivering stronger global search, faster convergence, and a better balance between exploration and exploitation.
Problem Description Develop a quantum-inspired metaheuristic optimization framework that dynamically generates near-optimal vehicle routes under real-time or simulated traffic conditions.The …

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109. [SIH26138] Quantum-Inspired Fuel Consumption Prediction and Green Fleet Optimization

Organization: Egreen Quanta | **Theme:** Smart Vehicles | **Submission Deadline:** 20 September 2026

Background The maritime and logistics industries are under increasing pressure to reduce greenhouse gas emissions while maintaining operational efficiency and cost-effectiveness. Fuel consumption constitutes one of the largest operational expenses and environmental impacts of fleet operations. Traditional optimization and prediction methods often struggle with the high-dimensional, non-linear, and multi-objective nature of green fleet management, especially when integrating alternative fuels, varying vessel types, and dynamic operational constraints.
Quantum-inspired metaheuristic algorithms offer a promising approach by combining the global search capabilities of quantum principles with classical computing, enabling more effective solutions for complex, large-scale fleet optimization problems.
Description This problem focuses on developing a quantum-inspired optimization and prediction …

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110. [SIH26139] Hybrid Quantum Machine Learning Platform for Early Disease Detection

Organization: Egreen Quanta | **Theme:** MedTech / BioTech / HealthTech | **Submission Deadline:** 20 September 2026

Background Early and accurate detection of diseases significantly improves treatment outcomes and reduces healthcare costs. Classical machine learning models have achieved notable success in medical diagnosis; however, they often face limitations when dealing with high-dimensional, noisy, and complex biomedical data (e.g., genomics, medical imaging, and electronic health records).
Quantum machine learning (QML) offers the potential to capture intricate patterns through quantum superposition and entanglement. Due to current hardware constraints, a hybrid quantum-classical approach provides a practical pathway to leverage quantum advantages while remaining executable on existing quantum simulators and near-term quantum devices.
Description This problem focuses on designing and developing a hybrid quantum machine learning platform for early disease detection. The platform will integrate …

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111. [SIH26140] AI-Based Interactive Quantum Algorithm Learning Platform

Organization: Egreen Quanta | **Theme:** Smart Education | **Submission Deadline:** 20 September 2026

Background Quantum computing is a transformative technology with significant impact across scientific and industrial domains. However, education in this field remains challenging due to the abstract nature of core concepts such as qubits, superposition, entanglement, and quantum algorithms.
Existing learning resources are often static, heavily theoretical, and lack hands-on interaction.
Limited access to real quantum hardware further restricts practical learning. There is a strong need for an integrated, interactive, and intelligent platform that combines theoretical instruction, visual circuit design, real-time simulation, and personalized AI-based guidance to accelerate quantum education and workforce development.
Description The goal is to develop an AI-powered interactive web-based platform that enables students, researchers, and professionals to learn, design, simulate, and …

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112. [SIH26141] Quantum-Inspired Cyber Threat Detection for Digital Signature Security

Organization: Egreen Quanta | **Theme:** Blockchain & Cybersecurity | **Submission Deadline:** 20 September 2026

Background The rapid advancement of quantum computing poses a serious threat to classical public-key cryptographic systems such as RSA and Elliptic Curve Cryptography (ECC), which can be broken by algorithms like Shor's algorithm. This vulnerability endangers the security of critical digital infrastructures. Quantum Digital Signature (QDS) protocols offer information-theoretic security by exploiting fundamental principles of quantum mechanics. Among these, teleportation-based QDS protocols are particularly promising because they enable secure signature generation and verification through quantum teleportation and entanglement, while reducing some of the practical deployment complexities associated with earlier QDS schemes.
Description This problem focuses on developing a quantum-inspired cyber threat detection framework specifically designed for Quantum Digital Signature (QDS) systems. …

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113. [SIH26142] Deep Learning Based Super Resolution Mapping (SRM) from Medium Resolution Satellite Imageries

Organization: National Technical Research Organisation (NTRO) | **Theme:** Smart Education | **Submission Deadline:** 20 September 2026

• Background Medium-resolution satellite imagery, typically ranging from 10 to 30 meters, is widely used in change detection, agriculture, land-cover mapping, disaster monitoring, and urban planning because it offers broad coverage and frequent revisit time. However, the spatial detail is often insufficient for fine-scale analysis, such as identifying small buildings, narrow roads, field boundaries, or localized damage assessment. This creates a need for advanced deep learning based generative enhancement techniques that can extract greater value from existing Earth observation data.
• Description Medium-resolution satellite imagery, usually ranging from 10 to 30 meters, is widely used in remote sensing for agriculture monitoring, land-cover mapping, urban planning, disaster assessment, and environmental observation because it provides large-area coverage and frequent revisit …

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114. [SIH26143] Leveraging satellite imagery to determine Oil spills at sea along with AIS data correlations to identify vessel responsible for the spill.

Organization: National Technical Research Organisation (NTRO) | **Theme:** Space Technology | **Submission Deadline:** 20 September 2026

• Background Marine oil spills inflict great damage on marine ecosystems and several times remains un-attributable to the vessel causing such spills. Leveraging satellite imagery along with AIS data will enable detection of oil spills and vessel responsible for the same.
• Description The core challenge attempts to facilitate detection of oil spills and also in identifying the polluting vessel using remote sensing satellite data, such as SAR and EO imagery and AIS data. Participants are to design an intelligent automated pipeline to do the following: (a) Detect and characterise the oil spill and calculating geometric properties and age if feasible. (b) Using oceanographic and meteorological data, it is envisaged to trace the slick towards the origin point and time, predict the future flow of the slick, and (c) analyse and attribute the spill to a vessel using historic AIS data to …

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115. [SIH26145] AI-Based Detection of Cyber Threats in Unidirectional IP Traffic

Organization: National Technical Research Organisation (NTRO) | **Theme:** Blockchain & Cybersecurity | **Submission Deadline:** 20 September 2026

• Background Critical-infrastructure operators observe their gateway and peering links using passive mirroring or hardware data diodes that copy traffic into a monitoring enclave in one direction only. The enclave can see everything crossing the link, but it has no physical or protocol-level path back into the production network. This is deliberate as it removes an entire class of attack in which a compromised monitoring or analytics system becomes a pivot into the core network, and it preserves a clean chain of custody for forensic use. The trade-off is that any intelligence layer sitting in that enclave must work purely from what it can passively observe such as packet captures, exported flow records (NetFlow/IPFIX/sFlow), and derived metadata with no ability to send probes, complete handshakes with the traffic source, or push a mitigation command back.
• Description The objective is …

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116. [SIH26146] AI-Powered Monitoring & Analysis of Bitcoin Transaction Traffic

Organization: National Technical Research Organisation (NTRO) | **Theme:** Transportation & Logistics | **Submission Deadline:** 20 September 2026

• Background Bitcoin's pseudonymous, peer-to-peer design lets criminal actors move, layer, and cash out illicit funds - ransomware payments, darknet-market proceeds, extortion, and laundering - while evading traditional financial surveillance.
The objective of problem statement is to design and build a complete system (offline) that ingests bulk Bitcoin transaction/network metadata (in CSV/JSON/XML), correlates network-layer (IP/port/timing) observations with blockchain-layer (wallet/TXID/amount) data, and applies AI/ML to detect anomalies, cluster entities, and generate prioritized, explainable investigative leads.
• Description i.Challenge Objectives- • Ingest & parse a bulk metadata dataset (timestamp, src/dst IP & port, TXID, input/output wallet addresses, amounts, fee, script type).
• Build an entity/transaction graph linking IPs, wallets, and transactions.
• Implement AI/ML …

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117. [SIH26147] Automated model for analysis of .IQ and .wav files along with signal parameter extraction

Organization: National Technical Research Organisation (NTRO) | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

• Background The raw data for analysis of signal collected off the air typically range from few Khz to Ghz bands. The analysis is being carried out manually to identify the signal parameters and the resultant data is then utilised for processing signals in the designated sensors. This data is often insufficient for fine grain analysis for parameter extraction such as modulation type, sampling rate, FEC, interleaving, etc. This creates a need for advanced data processing to extract the observation data.
• Description The terrestrial signals received from various sources includes data in HF, VHF and UHF bands. The raw data collected in the form of .wav or .IQ format to retain the characteristics of wave form. The analysis of signals is primarily dependent on the basic characteristics of data points selected during recording of these signals. Since the data point are recorded from …

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118. [SIH26148] Creation of scripts/functions with new programming language to commence Computer & Network forensic analysis without triggering security solutions

Organization: National Technical Research Organisation (NTRO) | **Theme:** Blockchain & Cybersecurity | **Submission Deadline:** 20 September 2026

• Background Modern antivirus solutions restrict proprietary software from executing or creating custom scripts designed to analyze the system for deep forensic system analysis. They rely heavily on behavioral heuristics, static signature matching, common compiler outputs (like standard MSVC or GCC artifacts), typical API call sequences and kernel-level monitoring to intercept activities. However, a significant paradigm shift may occur when programmers adopt sophisticated software engineering practices-specifically continuous integration and continuous deployment (CI/CD).
• Description Creating 'Next-Gen' programming language framework, named as 'JOCKY' using cross-platform compiler (windows & ubuntu) which enables systematic creation of scripts for analyzing malicious activities and also provide the complete digital forensics of the computer or network. By utilizing this specific new …

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119. [SIH26149] Design and Development of an Integrated Secure Data Erasure and Advanced File Recovery Tool for Digital Forensics and Data Sanitization

Organization: National Technical Research Organisation (NTRO) | **Theme:** Blockchain & Cybersecurity | **Submission Deadline:** 20 September 2026

• Background With the rapid growth of digital storage technologies, organizations, government agencies, law enforcement units, enterprises, and individual users face two major challenges: securely destroying sensitive data to prevent unauthorized recovery and recovering deleted digital evidence during forensic investigations. Existing solutions generally focus on either secure data deletion or file recovery and often support limited storage technologies and file systems. This forces investigators and cybersecurity professionals to use multiple tools, increasing complexity, cost, and operational inefficiencies. Therefore, there is a need for a unified platform that integrates secure data sanitization with advanced forensic-grade file recovery and carving capabilities.
• Description The proposed solution aims to develop an integrated software platform consisting of three core modules: (1) …

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120. [SIH26150] Development of a Multi-Vendor DVR/NVR Forensic Analysis Tool for Standardized Acquisition, Recovery, and Analysis of Surveillance Evidence.

Organization: National Technical Research Organisation (NTRO) | **Theme:** Blockchain & Cybersecurity | **Submission Deadline:** 20 September 2026

• Background Digital/Network Video Recorders (DVR/NVRs) are widely used for surveillance in government agencies, law enforcement, critical infrastructure, businesses, and residential environments. Major DVR/NVR manufacturers such as Dahua Technology, CP Plus, Honeywell Security, TP-Link, Godrej, Uniview, HIKVISON, and Matrix use proprietary storage formats, file systems, metadata structures, and video encoding mechanisms. During forensic investigations, surveillance footage serves as crucial digital evidence; however, the lack of standardization across DVR/NVR vendors makes acquisition, recovery, analysis, and validation difficult. Investigators often rely on multiple vendor-specific tools, resulting in increased investigation time, inconsistent results, timestamp synchronization issues, challenges in deleted footage recovery, and difficulties in maintaining evidence integrity. …

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121. [SIH26151] Dark web threat actor de-anonymization

Organization: National Technical Research Organisation (NTRO) | **Theme:** Blockchain & Cybersecurity | **Submission Deadline:** 20 September 2026

• Background The dark web has become a preferred operating space for threat actors in the modern age, mainly because it lets them hide their identity behind Tor hidden services, which makes attribution of threat actors operating on darkweb the main challenge for any investigation. Such threat actors carry out a wide range of unlawful activities such as drugs and arms sale, stolen data and hacking services, money laundering, terror financing, etc. The objective of this problem statement is to build a system for the deanonymization of dark web threat actors and link them to suspect real-world entities.
• Description The system shall deanonymize dark web threat actors by continuously gathering their footprints from a range of sources (marketplaces, forums, deep web etc.) and linking them to the identifying information available on those sources. The system envisages three core …

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122. [SIH26152] Social Media Analytics

Organization: National Technical Research Organisation (NTRO) | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

• Background Social media platforms are complex ecosystems driven by human emotion, diverse demographics, and interconnected networks. To truly understand an online community, it is required look beneath the surface. This requires understanding how followers feel (Sentiment Analysis), who those followers are (Demographics), what topics are captivating them (Trend Tracking), and how they influence one another (Link Analysis). Combining these four vectors using AI is the key to unlocking true audience intelligence.
• Description Participants will be challenged to design and build an AI-driven Social Media Analytics Framework that processes raw platform data to extract deep, actionable audience insights. The system must leverage advanced Artificial Intelligence and Machine Learning techniques to simultaneously infer follower sentiment, map audience demographics, identify top trending …

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123. [SIH26153] AI based Network Attack Forecasting from Network Traffic Data

Organization: National Technical Research Organisation (NTRO) | **Theme:** Blockchain & Cybersecurity | **Submission Deadline:** 20 September 2026

• Background This challenge seeks AI systems capable of learning network behaviour, anticipating attacker progression and supporting proactive cyber defence using the emerging concept of World Models. Design and develop a software prototype that learns the evolving state of a computer network from traffic telemetry and predicts the likelihood and progression of malicious activity before compromise is completed. The solution should ingest network traffic, learn temporal behaviour, forecast future attack states and provide interpretable decision support for defenders. Solutions should demonstrate applicability to enterprise environments and Critical Information Infrastructure.
• Represent network state using feature vectors or graphs.
• Learn state-transition dynamics using sequence models (LSTM, Transformer), Graph Neural Networks, latent state models or other AI techniques.
• Forecast …

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124. [SIH26154] Gen AI Platform for Automated Content Transformation

Organization: National Technical Research Organisation (NTRO) | **Theme:** Smart Automation | **Submission Deadline:** 20 September 2026

• Background Organisations frequently need to convert information available in different forms such as news articles, reports, advisories, threat intelligence, policy documents, research papers, announcements, incident reports or free-form prompts into specific communication artefacts suitable for various purposes. The process of manually analysing the source content, understanding the desired objective and creating the required output format is time-consuming, resource-intensive and often requires expertise in content creation, communication and domain knowledge.
There is a need for an intelligent platform that can transform user-provided content into a desired output format through a simple and configurable interface.
• Description The system shall act as an AI-powered content transformation engine that converts a common source of information into the specific deliverable requested by …

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125. [SIH26155] AI-Driven Multi-Vendor Network Security Compliance Auditor

Organization: National Technical Research Organisation (NTRO) | **Theme:** Blockchain & Cybersecurity | **Submission Deadline:** 20 September 2026

• Background Modern enterprise networks are inherently heterogeneous, consisting of a vast array of hardware from diverse vendors. Organizations are mandated to align these devices with rigorous security frameworks, including CIS Benchmarks, NIST SP 800-53, DISA STIGs, and ISO/IEC 27001.
The network environment includes, but is not limited to:
• Firewalls & SASE: Palo Alto, Fortinet, Cisco (Firepower/Secure/Meraki), Check Point, Juniper (SRX), Sophos, SonicWall, WatchGuard, Barracuda, Zscaler, Cloud-native firewalls (AWS, Azure, GCP), Sangfor, Hillstone, A10, Forcepoint, Stormshield, Netgate (pf/TNSR), Cato Networks, and others.
• Routers & Switches: Cisco (Catalyst/Nexus), HPE Aruba, Juniper (EX/MX/PTX), Arista, Extreme, NVIDIA (Mellanox), Allied Telesis, Huawei, D-Link, MikroTik, Ubiquiti, Alcatel-Lucent, Ruijie, Adtran, and others.
• Specialized Networking: Open/Disaggregated (Dell, …

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126. [SIH26156] Universal Log Pre-processing Framework

Organization: National Technical Research Organisation (NTRO) | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

• Background Modern enterprises generate massive volumes of logs from a wide range of sources, including network devices, servers, operating systems, applications, databases, cloud services, containers, endpoint security tools, identity and access management systems, IoT devices, and other hardware and software platforms. These logs are produced in diverse formats such as Syslog, JSON, XML, CSV, CEF, LEEF, proprietary vendor formats, and application-specific schemas.
The diversity of log structures creates significant challenges in centralized monitoring, security operations, compliance reporting, incident investigation, and threat analytics. Security teams often spend substantial effort developing source-specific parsers and normalization rules before the data can be effectively utilized by SIEM, data lake, or machine learning platforms.
As organizations adopt hybrid, multi-cloud, and …

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127. [SIH26157] Supervisory Analytics Tool for SOC Assessment (SAT-SA)

Organization: National Technical Research Organisation (NTRO) | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

Background:
The National Critical Information Infrastructure Protection Centre (NCIIPC) assesses the cyber resilience of Critical Sector Entities (CSEs).
As part of these assessments, NCIIPC performs manual reviews of samples of security alerts and case-management records generated by Security Operations Centres (SOCs). These reviews have consistently produced valuable supervisory findings that were not evident through policies, audits, self-assessments, management reports, KPI dashboards, or compliance documentation.
The purpose of these reviews is not to assess individual alerts. Rather, alert and case-management data are used as operational evidence to assess whether a CSE possesses effective capabilities relating to:
(i). Threat Detection (ii). Investigation (iii).Escalation (iv). Incident Response (v).Security Operations (vi).Governance and Oversight (vii).Operational Discipline …

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128. [SIH26158] Single-Pass Drone Video to Accurate 3D Model Generation System

Organization: National Technical Research Organisation (NTRO) | **Theme:** Robotics and Drones | **Submission Deadline:** 20 September 2026

• Background:
Generation of accurate 3D models of buildings, infrastructure, terrain, and objects typically requires multiple drone passes, extensive image overlap, specialized flight planning, and significant post-processing time. In operational scenarios such as disaster response, surveillance, infrastructure inspection, military reconnaissance, and rapid mapping, there is often only a single opportunity to capture data over the target area. A solution capable of generating an accurate and textured 3D model from a single drone pass video would significantly reduce mission time, operator effort, data acquisition requirements, and processing complexity while enabling near real-time situational awareness.
• Description:
Design and develop an AI-enabled system capable of generating a georeferenced and metrically accurate 3D model of a scene using only a single-pass drone video stream …

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129. [SIH26159] SecureMailScope: AI-Assisted Cryptographic Security Posture Assessment for Secure Email Communications

Organization: National Technical Research Organisation (NTRO) | **Theme:** Blockchain & Cybersecurity | **Submission Deadline:** 20 September 2026

• Background Electronic mail remains one of the most critical communication services for governments, enterprises, financial institutions, and academic organizations. Despite the widespread adoption of Transport Layer Security (TLS), many SMTP, IMAP, and POP3 deployments continue to suffer from cryptographic misconfigurations such as obsolete TLS versions, weak cipher suites, insecure STARTTLS implementations, expired or improperly configured certificates, and non-compliance with modern security standards. These weaknesses expose email infrastructures to downgrade attacks, man-in-the-middle attacks, passive interception, and other cryptographic threats.
Although existing network analysis tools provide extensive packet-level visibility, they primarily focus on protocol decoding and traffic inspection. They do not automatically evaluate the overall cryptographic security posture of email …

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130. [SIH26160] AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework

Organization: National Technical Research Organisation (NTRO) | **Theme:** Blockchain & Cybersecurity | **Submission Deadline:** 20 September 2026

• Background Virtual Private Networks (VPNs) are fundamental to secure communication over untrusted networks. Among the available VPN technologies, IPsec is widely adopted across enterprise, government, military, and cloud infrastructures because of its ability to provide confidentiality, integrity and authentication.
However, the security of an IPsec deployment depends on multiple factors, including the chosen cryptographic algorithms, authentication mechanisms, key exchange protocols, and operational mode (Tunnel or Transport). Misconfigurations, outdated cipher suites, improper key management, or protocol implementation flaws can significantly weaken the overall security posture.
Traditional protocol analysis tools provide packet-level visibility but often require expert interpretation. There is a growing need for intelligent systems capable of automatically analyzing IPsec …

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131. [SIH26161] Dam Break Inundation Modelling Using Hydrodynamic Modelling of any River

Organization: National Technical Research Organisation (NTRO) | **Theme:** Disaster Management | **Submission Deadline:** 20 September 2026

• Background In India, due to natural disaster various natural dam / lake formations were observed which can be a major reason of flash flood in the lower catchment, for example, natural lake formed over the Rishi Ganga river of Uttarakhand in Feb 2021, Wapriyang river in Nov 2021, Phuktal river near Sumdo, J&K in Mar 15, Kosi river in 2008 etc. Devastating flood happened in the Kashmir valley, Assam in 2014 and many other places over a period of time. Therefore, simulation modelling for flash flood and scenario generation is important from Humanitarian Assistance and Disaster Relief (HADR) point of view. Another important aspect is water release issues from the Dam of major rivers. In the crisis situation, if the dam brakes, how much water will flow into the river and what are the area it will inundated / impacted need to be estimated. In order to carry out this work simulation …

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132. [SIH26162] AI-Based Detection and Classification of Industrial Fires and Persistent Thermal Sources Using NASA FIRMS, OSM & Satellite Data

Organization: National Technical Research Organisation (NTRO) | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

• Background Industrial facilities generate thermal signatures that can be observed from space, but current satellite-based monitoring systems like NASA FIRMS cannot distinguish between different types of thermal anomalies. To address this, there is a challenge to develop an AI-enabled geospatial system that integrates thermal data, land-cover information, industrial databases, and satellite imagery to automatically identify, classify, and monitor industrial fires and persistent thermal sources.
• Description Industrial facilities such as oil refineries, petrochemical complexes, thermal power plants, steel industries, mining areas, and LNG terminals generate thermal signatures that can be observed from space. In addition, accidental industrial fires, gas leaks, explosions, and abnormal thermal events pose significant risks to critical infrastructure, public safety, and the …

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133. [SIH26163] Security Assessment of the World Monitor application

Organization: National Technical Research Organisation (NTRO) | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

• Background The world Monitor application is a Web/ Mobile platform that provides users with real-time monitoring, analytics, and reporting features. The application handles user authentication, data visualization, API communication, and role-based access controls.
As a security analyst, the task is to evaluate the application's security posture and identify vulnerabilities that could compromise the confidentiality, integrity, or availability of the system.
• Description Conduct an authorized security assessment of the World Monitor application to:
1. Identify security vulnerabilities in the application.
2. Assess the potential impact of each vulnerability.
3. Demonstrate proof-of-concept exploitation in a controlled environment.
4. Recommend remediation measures to mitigate the identified risks.
• Scope The assessment should focus on:
• Authentication and session management
• …

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134. [SIH26164] Enterprise Cryptographic Discovery & Analysis Tool (ECDAT)

Organization: National Technical Research Organisation (NTRO) | **Theme:** Blockchain & Cybersecurity | **Submission Deadline:** 20 September 2026

• Background Transitioning to Post Quantum Cryptography based solutions requires preparedness, risk assessment and financial and operational investment. Towards this, discovery and inventory of Cryptographic Artefacts is the critical first step, that will enable the transition.
• Description i. Identify and catalogue all cryptographic artefacts (algorithms, keys, certificates, protocols, libraries, hardware modules, cloud services) across internal and external facing applications, products and infrastructure.
ii. The tool should perform a comprehensive quantum risk assessment and identify systems prone to potential quantum attacks, and highlight risks to sensitive data.
iii. Classify all the artefacts by type, lifetime and business criticality. Apply structured frameworks such as Mosca’s algorithm (compare data lifetime plus migration time against expected arrival of cryptographic …

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135. [SIH26165] AI/NLP Engine to Detect Serious Injury & Fatality (SIF) Precursors in OIL's Unsafe-Act/Unsafe-Condition and Near-Miss Reports

Organization: Oil India Limited | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

• Background OIL collects large volumes of UA/UC observations, near-miss and incident reports through its HSSE platform but these are triaged manually after certain time intervals such as monthly, quarterly etc.However, Global best practice (DEKRA Martin & Black 2015; EEI SIF Precursor model; VelocityEHS 2024 PSIF classifier) has established that low-severity incidents do not share the same causes as fatalities - non-fatal US accidents fell 51% over 15 years while fatalities fell only 25.5%.Leading operators therefore separately flag the ~20–25% of reports carrying genuine fatal potential.
Problem Description Build a prototype that ingests OIL's free-text safety reports and automatically a) Classifies each as SIF-potential vs non-SIF-potential b) Tags it to the relevant IOGP Life-Saving Rule (e.g., Energy Isolation, Hot Work,Confined Space, Line of Fire)
c) Surfaces recurring precursor …

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136. [SIH26166] Multi-modal, Sun angle and scale invariant image correspondence using Chandrayaan-2 optical images (OHRC, TMC and IIRS)

Organization: Indian Space Research Organisation(ISRO) | **Theme:** Space Technology | **Submission Deadline:** 20 September 2026

Background Image Registration is the process of aligning two or more images of the same scene taken at different times, from different viewpoints, or by different sensors into a common coordinate system.
It has two main components:
• Source Image (Moving): The image that is to be geometrically transformed to align with the reference image.
• Reference Image (Fixed): The target image about which source image is to be geometrically transformed.
Description The process of lunar images registration involves finding match points between source and reference image and then aligning the source image with the reference image. The key challenges involved in this process are as follows:
• Illumination variation: Illumination variation refers to changes in sun azimuth and elevation effect on the surface lighting conditions that affect the appearance of the lunar surface features which is hard to …

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137. [SIH26167] SatQuery AI - An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries

Organization: Indian Space Research Organisation(ISRO) | **Theme:** Space Technology | **Submission Deadline:** 20 September 2026

Background Remote-sensing imagery is widely used for agricultural monitoring, disaster management, urban planning, forest monitoring, water-resource assessment, infrastructure mapping, and environmental analysis. However, most existing remote-sensing AI solutions are developed as isolated applications for a single predefined task, such as land-cover classification, object detection, visual question answering, or change detection. These systems often require users to understand satellite-data characteristics, GIS workflows, model selection, and task-specific parameters. Consequently, non-expert users may find it difficult to obtain meaningful information from satellite imagery through simple natural-language queries.
Many operational remote-sensing questions cannot always be answered reliably using a single optical image. Relevant information may be distributed across paired or multiple …

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138. [SIH26168] AI-ML based Intelligent Dead Reckoning system for seamless navigation

Organization: Indian Space Research Organisation(ISRO) | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

Background Vehicle logistics, ride-hailing services, quick commerce and emergency responders heavily rely on smartphone-based navigation apps (like Google Maps or MapmyIndia) powered by GNSS (GPS/Galileo/NavIC etc). However, when a vehicle enters a long underground tunnel/ underpass, a multi-level parking lot, a dense forested highway, or a deep urban canyon surrounded by skyscrapers, GNSS connectivity drops entirely. GNSS signals are inherently weak and vulnerable to structural blockage (urban canyons, dense foliage, tunnels, deep valleys) and unintentional electromagnetic interferences from variety of sources such as jamming. This causes navigation apps to freeze, jump erratically, or miscalculate upcoming turns, leading to missed exits, delivery delays, and safety hazards.
In these environments, systems must rely on self-contained Inertial Navigation Systems (INS) built from Inertial …

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139. [SIH26169] Development of an AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile Free Space Optical Communication (FSOC) Terminals

Organization: Indian Space Research Organisation(ISRO) | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

Background Free Space Optical Communication (FSOC) offers unprecedented advantages for next-generation mobile networks, including gigabit-to-terabit data rates, license-free spectrum operation, high immunity to electromagnetic interference, etc. However, deploying FSOC links between mobile platforms (satellites, UAVs presents a severe challenge of pointing, acquisition and tracking (PAT) of highly narrow laser beams. PAT typically happens in two stages: coarse alignment and fine alignment. Coarse alignment is one of the key challenges of PAT, where the transmitting terminal must first locate and maintain the remote terminal within its camera Field-of-View (FOV).
Developing and testing such algorithms on real hardware requires expensive cameras, pan-tilt mechanisms, and optical components & equipment. A software based virtual camera tracking provides an inexpensive and accessible …

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140. [SIH26170] AI-Driven Anomaly Detection in Component Burn-In & Screening

Organization: Indian Space Research Organisation(ISRO) | **Theme:** Smart Automation | **Submission Deadline:** 20 September 2026

Background In high-reliability sectors (like space) electronic components undergo rigorous environmental stress screening (ESS), including Burn-In testing (operating components at elevated temperatures, e.g., 125°C for extended periods).
Traditional screening relies on static parametric pass/fail limits. However, 'latent defects'-components that pass the absolute limits but exhibit subtle, anomalous drift over time-often escape into final payloads, leading to catastrophic field failures.
Description Development of a predictive machine learning model that analyzes time-series parametric data (e.g., standby current Iddq, leakage currents, or propagation delays measured at intervals like 0h, 24h, 96h, and 168h to detect anomalous components.
Expected Solution Module A: The outlier detection system Static limits catch obvious failures. Participants need to develop a 'Dynamic' outlier …

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141. [SIH26171] On-device Visual Perception for Light-weight Browser Agents

Organization: Indian Space Research Organisation(ISRO) | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

Background AI agents are becoming omnipresent in the current era and can play an important role in our digital interactions. If an agentic AI pipeline has access to our visual context, screen states, they can assist users in complex workflows and automate many tasks. Most of the agentic AI pipelines are deployed on server side which limits the type to data that a user can share with it. It would open a new dimension of possibilities, if a local agent is deployed on user machine particularly browser which can eliminate the need to share the sensitive data with the server. Local system generally has fewer resources than server and is unable to host a full-fledged pipeline therefore only the non-sensitive data such as structure of the screen, application fields etc can be sent to server for processing.
Modern browser APIs (such as WebGPU and WebAssembly) and local inference libraries (like …

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142. [SIH26173] iTantra -Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for low bitrate links

Organization: Indian Space Research Organisation(ISRO) | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

Background As vocal audio information is very data intensive making it difficult to transmit through low data rate links. In alert and distress based scenarios Transmitting Audio information is critical instead of written message as it will be more inclusive and will cater to everyone even if they are literate or not.
Description Build an Android App with lightweight, highly accurate STT and TTS models for 10 Indian Languages (Hindi, Gujarati, Marathi, Kannada, Malayalam, Tamil, Telugu, Odia, Bengali, English) that runs locally on a low-power device. The system’s STT module when activated after detecting pauses and stoppages should form the sentences detected and must instantly and efficiently stream the data through wifi/Bluetooth connected embedded device or another phone with same application with minimal latency. The systems TTS module when activated after receiving the Text data …

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143. [SIH26174] AI Human Activity Recognition for On-board BAS Experiments

Organization: Indian Space Research Organisation(ISRO) | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

Background As humanity aims for space missions such as BAS and lunar missions, real-time ground support becomes impossible due to communication delays. An AI-based HAR system acts as an on-board assistant that supports the execution of scientific experiments, ensuring the success of science beyond Earth's orbit.
In the space environment, AI-based HAR system may act as mission-critical support for astronauts. By tracking astronaut movements and activities in real time, HAR ensures scientific experiments and related protocols are executed flawlessly without requiring constant, high-bandwidth communication with mission control.
Description Challenge is to design and train an AI model that recognizes and validates the sequence of a pre-defined experiment using human activity recognition techniques.
Standalone operation: Space stations operate on restricted data bandwidth to Earth. Rather …

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144. [SIH26175] DepthWizard - Single-View Height Estimation and 3D Flythrough

Organization: Indian Space Research Organisation(ISRO) | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

Background Accurate Digital Elevation Models (DEMs) and Digital Surface Models (DSMs) are fundamental to urban planning, disaster management, and military reconnaissance. Traditionally, elevation data is acquired through stereo-imaging pairs, LiDAR, or Interferometric Synthetic Aperture Radar (InSAR). These approaches can be cost-prohibitive, dependent on specific sensor availability, and computationally intensive. Single-view height estimation offers an agile alternative, but foundational monocular depth models are trained largely on natural egocentric imagery and predict relative depth. When applied to remote sensing, they face domain gaps, structural variations, and a lack of absolute-scale mapping. Converting relative depth into metric elevation remains a critical challenge, alongside the operational need to transform static elevation profiles into interactive 3D assets that can be …

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145. [SIH26176] ORCA Marine EcOsystem Reasoning with Collaborative Agents

Organization: Indian Space Research Organisation(ISRO) | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

Background The marine ecosystem plays a vital role in supporting livelihoods, food security, biodiversity, maritime transportation, coastal resilience, and the blue economy. Every day, vast volumes of satellite Earth Observation and oceanographic data, including Sea Surface Temperature (SST), chlorophyll concentration, and weather forecasts, are generated by ISRO and other global agencies.
Marine stakeholders such as fishermen, researchers, coastal authorities, disaster management agencies, and maritime operators rely on timely access to oceanographic and meteorological information for operational planning and decision-making. As the volume, diversity, and complexity of marine data continue to increase, there is a growing need for an intelligent conversational platform that enables users to interact naturally with marine information, ask questions, explore scenarios, and receive …

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146. [SIH26182] Automated Attribution of Unknown Cryptocurrency Wallets to Nearest Virtual Asset Service Providers (VASPs) through Blockchain Intelligence APIs

Organization: Ministry of Home Affairs | **Theme:** Blockchain & Cybersecurity | **Submission Deadline:** 20 September 2026

• Background The rapid adoption of Virtual Digital Assets (VDAs) and decentralized blockchain ecosystems has significantly increased the complexity of cybercrime investigations globally. Law Enforcement Agencies (LEAs) frequently encounter cryptocurrency wallet addresses linked to cyber frauds, ransomware, investment scams, darknet activities, and laundering of crime proceeds.
Under the existing investigation workflow, LEAs raise lawful information disclosure requests through the SAHYOG Portal to Virtual Asset Service Providers (VASPs) such as crypto exchanges, custodial wallet providers, and trading platforms. However, in many cases, the suspect wallet identified during investigations belongs to an unhosted wallet or a wallet for which the associated VASP is unknown. This creates major delays in attribution, freezing of assets, and identification of the beneficial owner.
Blockchain …

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147. [SIH26183] Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics

Organization: Ministry of Home Affairs | **Theme:** Blockchain & Cybersecurity | **Submission Deadline:** 20 September 2026

• Background Cyber fraud victims increasingly report suspect cryptocurrency wallet addresses used by fraudsters for collection of funds in cases involving:
• investment scams,
• task-based frauds,
• sextortion,
• ransomware,
• phishing,
• darknet transactions,
• and organized cyber-enabled financial crimes.
During investigations, the reported wallet addresses are often:
• non-custodial wallets,
• temporary burner wallets,
• or intermediary wallets used for layering and laundering.
The inability to quickly identify the cryptocurrency exchange or VASP associated with these wallets delays:
• freezing of assets,
• preservation of evidence,
• tracing of fund flows,
• and victim fund recovery.
Manual blockchain tracing requires significant technical expertise and time, particularly in cases involving:
• multi-chain transfers,
• DeFi protocols,
• mixers/tumblers,
• bridges,
• and …

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148. [SIH26184] Development of a Predictive Analytics Framework for Cybercrime Complaints to Forecast Likely Cash Withdrawal Locations in Advance, Enabling Generation of Actionable Intelligence for Timely and Proactive Cybercrime Intervention.

Organization: Ministry of Home Affairs | **Theme:** Blockchain & Cybersecurity | **Submission Deadline:** 20 September 2026

• Background The National Cybercrime Reporting Portal is the centralized Portal, which is serving the whole country. Currently, the Portal facilitates citizens in filing complaints, LEAs act on complaints, Banking/Financial Institutions for their actions along with reports/graphs being pulled on daily basis. Presently, the Portal is receiving approximately 8000 complaints on daily basis. The number of complaints has increased manifold during the past months, and this will continue to rise in future. To address the issue of increasing cybercrimes, the proactive approach shall be adopted.
• Description This framework focuses on the mitigation of cybercrimes by adopting a proactive approach. The framework's output will enable the prediction of likely cash withdrawal locations, which, in turn, will allow law enforcement agencies (LEAs)
at the state and local levels, coordinated by I4C, to …

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149. [SIH26186] AI-Based Predictive Personnel Stress and Welfare Monitoring System for Uniformed Forces

Organization: Ministry of Home Affairs | **Theme:** MedTech / BioTech / HealthTech | **Submission Deadline:** 20 September 2026

• Background Personnel serving in Central Armed Police Forces (CAPFs), Armed Forces, and other uniformed services operate under physically demanding, psychologically stressful, and often hazardous conditions.Extended deployments, operational pressures, separation from families,irregular working hours, and exposure to traumatic incidents can significantly impact mental well-being.Currently, stress identification largely depends on manual observation and self-reporting, which may delay timely intervention. There is a need for a proactive, technology-driven solution that can identify early indicators of stress, burnout, and psychological distress while maintaining privacy and organizational trust.
• Description The proposed solution aims to develop an AI-powered Personnel Stress and Welfare Monitoring System capable of identifying potential indicators of stress, burnout, emotional fatigue, …

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150. [SIH26187] AI-Based Intelligent Video Analytics Platform for Border Surveillance using existing CCTV Infrastructure.

Organization: Ministry of Home Affairs | **Theme:** Smart Automation | **Submission Deadline:** 20 September 2026

• Background Border security forces deploy CCTV cameras at Border Out Posts(BOPs), check posts, border roads, and other strategic locations for surveillance and monitoring. However, conventional CCTV systems primarily provide video recording and live monitoring capabilities,requiring continuous human observation. Advanced surveillance functionalities such as Facial Recognition Systems (FRS), Automatic Number Plate Recognition (ANPR), intrusion detection, and object tracking often require specialized hardware and proprietary solutions,making large-scale deployment costly and difficult, particularly in remote border areas.
• Description The proposed solution aims to develop an AI-driven software platform capable of transforming existing CCTV infrastructure into an intelligent surveillance network without requiring dedicated FRS, ANPR, or smart-camera hardware. The platform shall ingest …

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151. [SIH26188] AI-Based Fake Identity & Document Screening System

Organization: Ministry of Home Affairs | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

• Background Common challenges faced at border checkpoints:
• Fake passports and visas
• Altered photographs
• Modified dates of birth
• Tampered visa stamps
• Identity impersonation
• Multiple identities used by the same person Expired or blacklisted travel documents
• High passenger volume causing delays Current verification methods rely heavily on human inspection and basic database lookups.
• Detailed Description Border checkpoints process thousands of identity documents every day,including passports, visas, national identity cards, permits, and travel authorizations. Manual verification is time-consuming, prone to human error, and often unable to detect sophisticated forgeries, tampering, or identity fraud.Develop an AI-powered document screening platform that automatically analyzes identity and travel documents, detects signs of tampering or forgery, validates information against …

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152. [SIH26189] AI-Powered Criminal Network Analysis System

Organization: Ministry of Home Affairs | **Theme:** Blockchain & Cybersecurity | **Submission Deadline:** 20 September 2026

• Background Modern criminal activities are increasingly organized and interconnected. Criminals often operate through networks involving associates, intermediaries, financial channels, communication links,locations, and events. Law enforcement agencies collect large volumes of data from sources such as:
• FIRs and police reports
• Call Detail Records (CDRs)
• Financial transaction records
• Surveillance reports
• Social media intelligence
• Criminal history databases
• Intelligence agency reports Despite having access to this information, investigators frequently face challenges in identifying hidden relationships among suspects because the data is fragmented, unstructured, and distributed across multiple systems. Manual analysis can be slow, labor-intensive, and prone to missing critical connections.With advances in Artificial Intelligence (AI), Machine Learning (ML),Natural Language …

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153. [SIH26190] Secure Digital Document Management System for Legal and Investigation Documents

Organization: Ministry of Home Affairs | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

• Background Law enforcement agencies, courts, legal departments, and investigative organizations handle vast amounts of sensitive documents throughout the lifecycle of a case. These documents may include:
• FIRs and police reports
• Investigation records
• Witness statements
• Charge sheets
• Court filings
• Evidence records
• Forensic reports
• Legal notices and judgments Many organizations still rely on paper-based systems or fragmented digital storage solutions. This often leads to challenges such as:
• Difficulty in locating documents quickly
• Unauthorized access to confidential information
• Document tampering risks
• Lack of version control
• Inefficient collaboration between departments
• Delays in legal and investigative processes
• Poor auditability and compliance tracking As the volume of legal and investigation-related data continues to grow,there is an increasing need for …

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154. [SIH26191] Intelligent Identification of Hazard-Based Red Zones, Carrying Capacity Assessment, and Immediate Relocation Needs for Vulnerable Habitations

Organization: Ministry of Home Affairs | **Theme:** Disaster Management | **Submission Deadline:** 20 September 2026

• Background India's disaster-prone regions face recurring hazards such as landslides,floods, coastal erosion, and cloudbursts. Vulnerable habitations often remain in unsafe zones, leading to repeated loss of lives and property.Current relocation efforts are largely reactive, initiated after disasters strike, rather than proactively planned.
• Description The initiative seeks to develop an intelligent, GIS-enabled decision support platform. This platform will dynamically identify and update multi-hazard Red Zones (areas unsuitable for permanent habitation),assess the carrying capacity of safer alternative sites, and prioritize vulnerable habitations for relocation. The system will integrate hazard intensity, population vulnerability, and disaster history to guide evidence-based decisions.
• Expected Solution A robust, AI-driven GIS platform that Maps and updates hazard-based Red Zones …

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155. [SIH26192] Flash Flood Prediction System for Hilly Regions using Multi-Source Data Theme

Organization: Ministry of Home Affairs | **Theme:** Disaster Management | **Submission Deadline:** 20 September 2026

• Background Hilly states in India are highly vulnerable to landslides and flash floods,which often occur with very short warning times. These sudden events result in significant loss of lives and property, and current early warning mechanisms are inadequate for hyper-local prediction and timely evacuation.
• Description The proposed initiative aims to develop a predictive system that integrates multiple data sources - rainfall data, soil moisture sensors,slope stability models, historical landslide inventories, and real-time IoT inputs. By combining these datasets, the system will generate hyper-local forecasts at the village or ward level, providing sufficient lead time for evacuation and risk mitigation.
• Expected Solution A comprehensive flash flood prediction system that Integrates rainfall,soil moisture, slope stability, and historical disaster data, utilizes IoT sensors for …

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156. [SIH26193] Student Innovation

Organization: AICTE | **Theme:** Smart Resource Conservation | **Submission Deadline:** 20 September 2026

Ideas focused on the intelligent use of resources for transforming and advancements of technology with combining the artificial intelligence to explore more various sources and get valuable insights.

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157. [SIH26194] Student Innovation

Organization: AICTE | **Theme:** Fitness & Sports | **Submission Deadline:** 20 September 2026

Ideas that can boost fitness activities and assist in keeping fit.

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158. [SIH26195] Student Innovation

Organization: AICTE | **Theme:** Heritage & Culture | **Submission Deadline:** 20 September 2026

Ideas that showcase the rich cultural heritage and traditions of India.

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159. [SIH26196] Student Innovation

Organization: AICTE | **Theme:** MedTech / BioTech / HealthTech | **Submission Deadline:** 20 September 2026

Cutting-edge technology in these sectors continues to be in demand. Recent shifts in healthcare trends, growing populations also present an array of opportunities for innovation.

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160. [SIH26197] Student Innovation

Organization: AICTE | **Theme:** Agriculture, FoodTech & Rural Development | **Submission Deadline:** 20 September 2026

Developing solutions, keeping in mind the need to enhance the primary sector of India - Agriculture and to manage and process our agriculture produce.

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161. [SIH26198] Student Innovation

Organization: AICTE | **Theme:** Transportation & Logistics | **Submission Deadline:** 20 September 2026

Creating intelligent devices to improve commutation sector.

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162. [SIH26199] Student Innovation

Organization: AICTE | **Theme:** Fitness & Sports | **Submission Deadline:** 20 September 2026

Submit your ideas to address the growing pressures on the city’s resources, transport networks, and logistic infrastructure.

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163. [SIH26200] Student Innovation

Organization: AICTE | **Theme:** MedTech / BioTech / HealthTech | **Submission Deadline:** 20 September 2026

There is a need to design drones and robots that can solve some of the pressing challenges of India such as handling medical emergencies, search and rescue operations, etc.

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164. [SIH26201] Student Innovation

Organization: AICTE | **Theme:** Smart Resource Conservation | **Submission Deadline:** 20 September 2026

Solutions could be in the form of waste segregation, disposal, and improve sanitization system.

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165. [SIH26202] Student Innovation

Organization: AICTE | **Theme:** Travel & Tourism | **Submission Deadline:** 20 September 2026

A solution/idea that can boost the current situation of the tourism industries including hotels, travel and others.

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166. [SIH26203] Student Innovation

Organization: AICTE | **Theme:** Renewable / Sustainable Energy | **Submission Deadline:** 20 September 2026

Innovative ideas that help manage and generate renewable /sustainable sources more efficiently.

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167. [SIH26204] Student Innovation

Organization: AICTE | **Theme:** Miscellaneous | **Submission Deadline:** 20 September 2026

Provide ideas in a decentralized and distributed ledger technology used to store digital information that powers cryptocurrencies and NFTs and can radically change multiple sectors.

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168. [SIH26205] Student Innovation

Organization: AICTE | **Theme:** Smart Education | **Submission Deadline:** 20 September 2026

Smart education, a concept that describes learning in digital age. It enables learners to learn more effectively, efficiently, flexibly and comfortably.

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169. [SIH26206] Student Innovation

Organization: AICTE | **Theme:** Disaster Management | **Submission Deadline:** 20 September 2026

Disaster management includes ideas related to risk mitigation, Planning and management before, after or during a disaster.

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170. [SIH26207] Student Innovation

Organization: AICTE | **Theme:** Travel & Tourism | **Submission Deadline:** 20 September 2026

Technology ideas in tertiary sectors like Hospitality, Financial Services, Entertainment and Retail.

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171. [SIH26208] Student Innovation

Organization: AICTE | **Theme:** Heritage & Culture | **Submission Deadline:** 20 September 2026

Challenge your creative mind to conceptualize and develop unique toys and games based on our civilization, history, and culture etc.

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172. [SIH26209] Student Innovation

Organization: AICTE | **Theme:** Space Technology | **Submission Deadline:** 20 September 2026

Space technology refers to the application of engineering principles to the design, development, manufacture, and operation of devices and systems for space travel and exploration.
